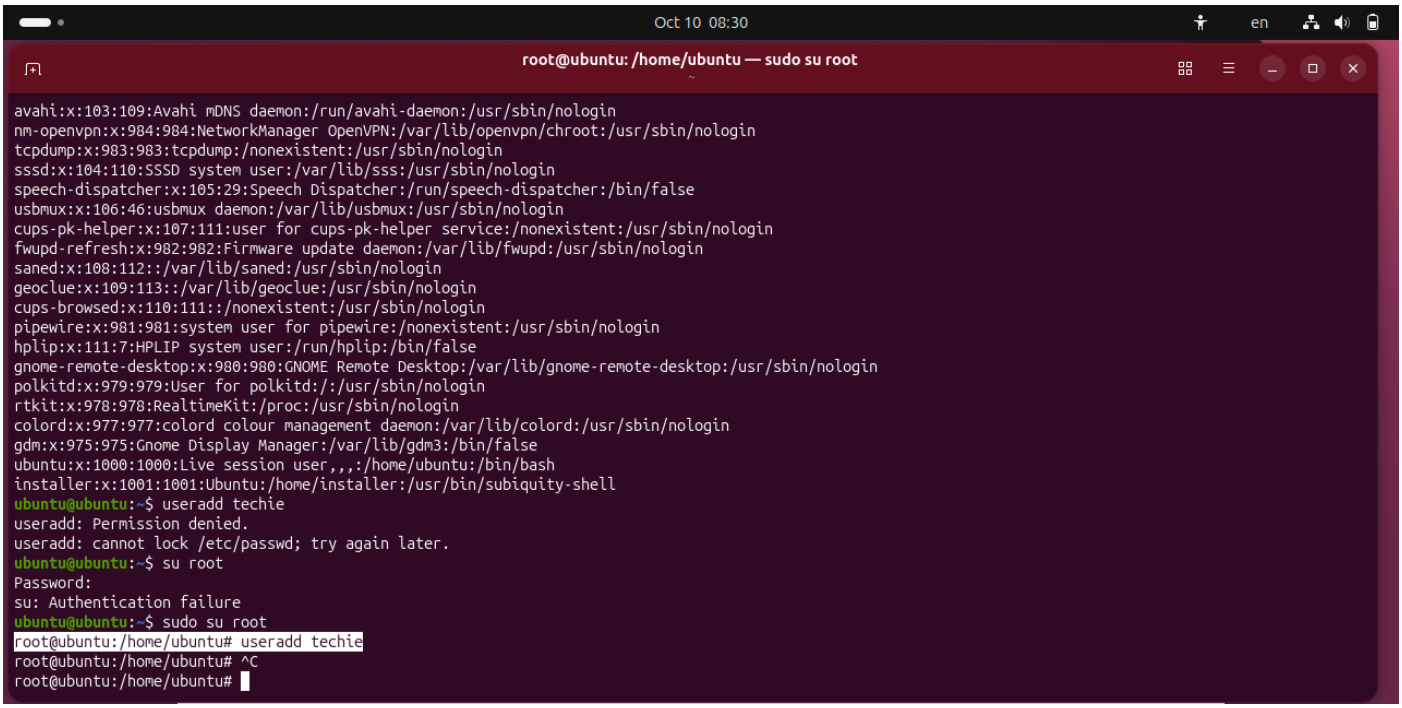
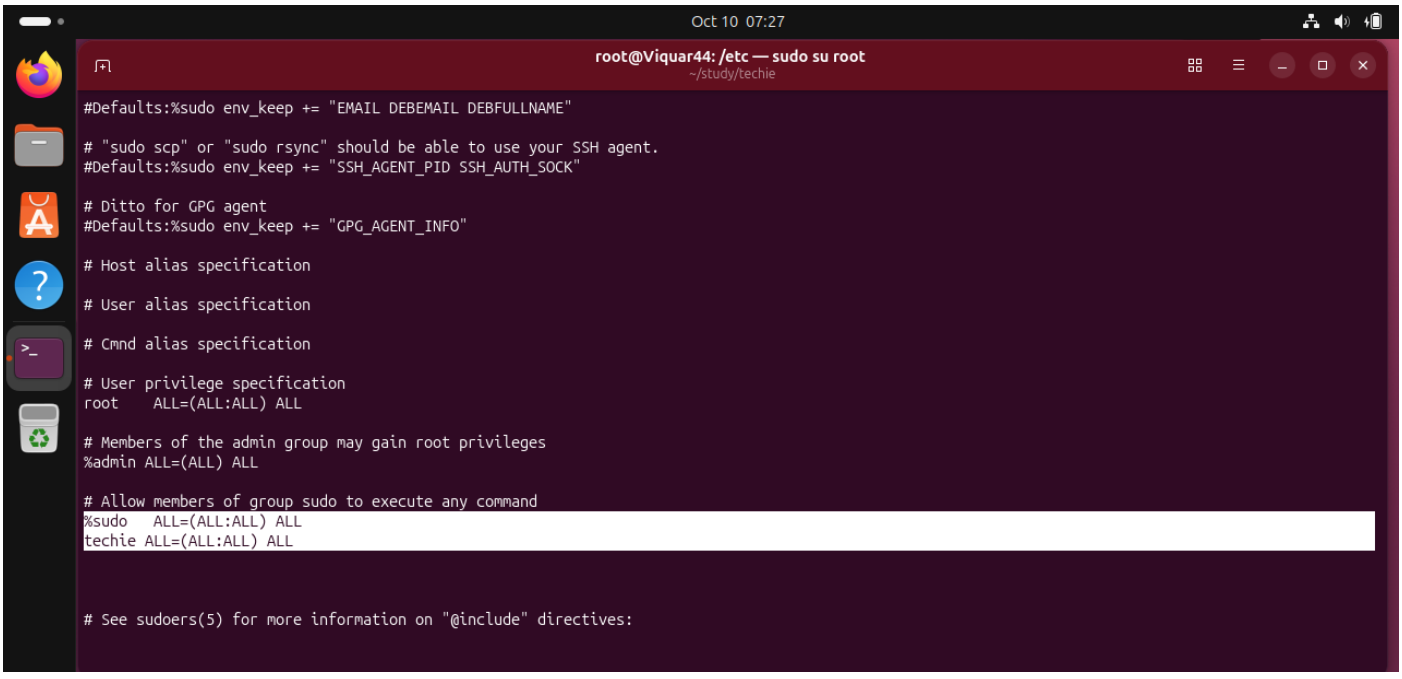
Linux Tasks

1. **Create user with name Techie and provide sudo access to user.**
   created a user (techie) by using command useradd and give all permissions to user (techie) by using method of changing sudoers file (techie ALL=(ALL:ALL) ALL
   edited sudoers file with VI editor

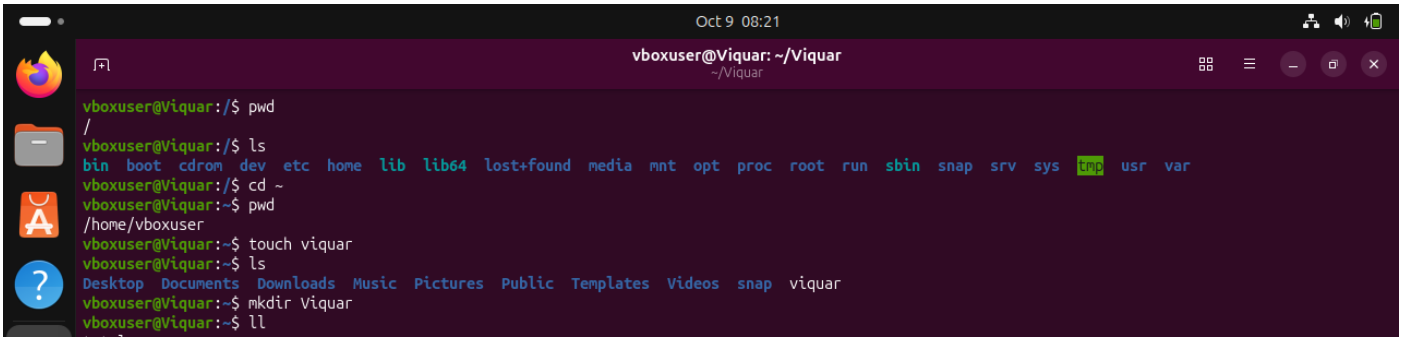




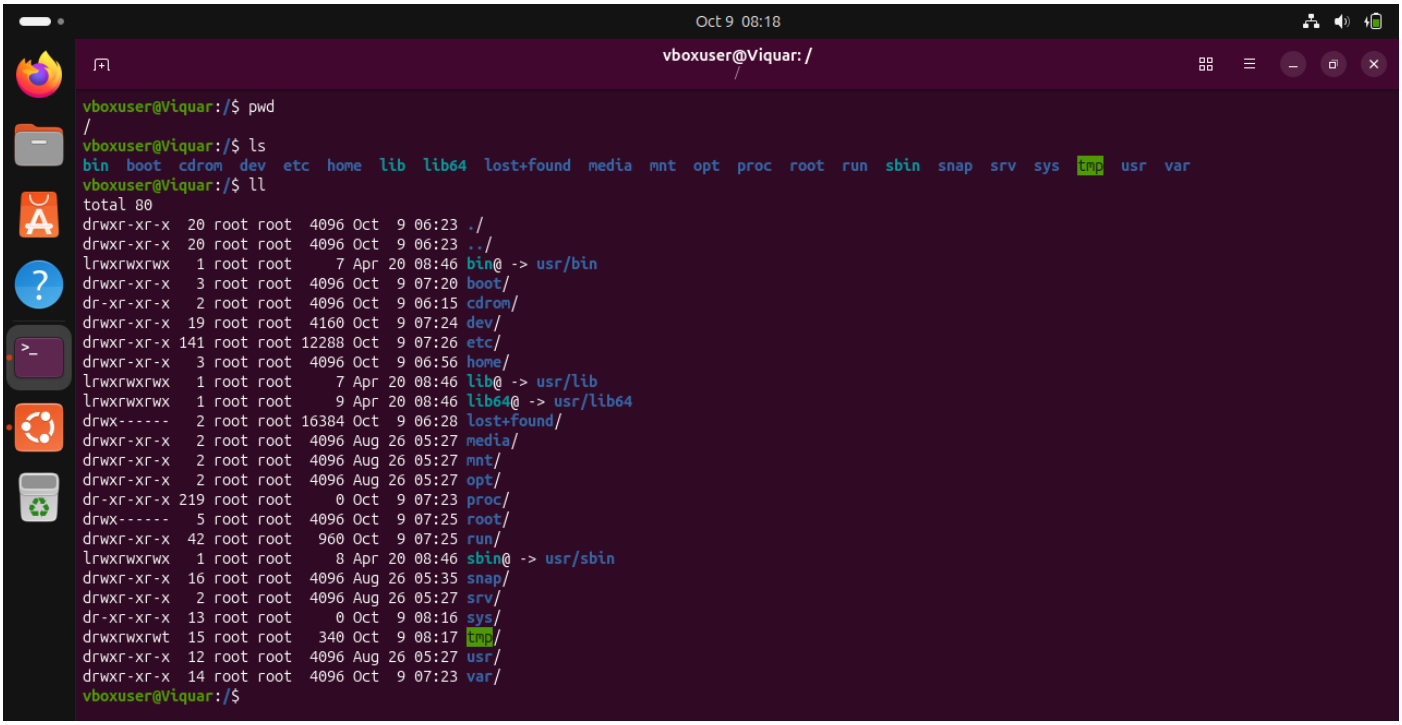
2. **Navigate to the home directory.**
   Used cd ~ command to navigate home directory



3. **Create a new directory.**
   used mkdir to create new directory



4. **List the contents of a directory.**
   used ls and ll command | ls (short list) ll (long list)



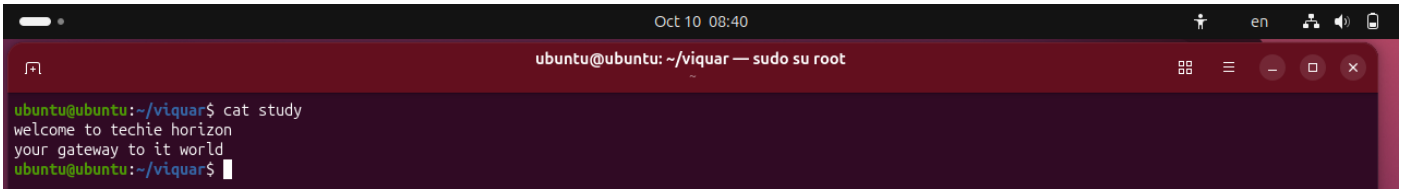
5. **Change the current directory.**
   used cd / to change directory from home to user



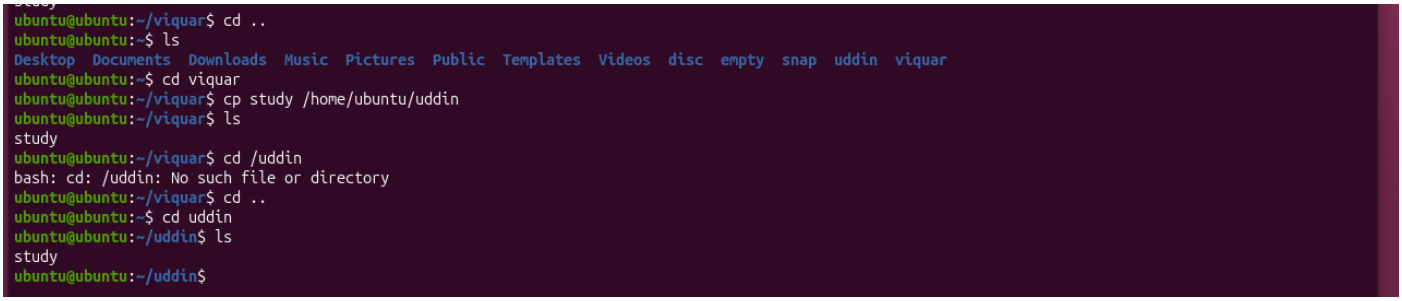
6. **Create a new empty file.**
   used touch command to create new empty file



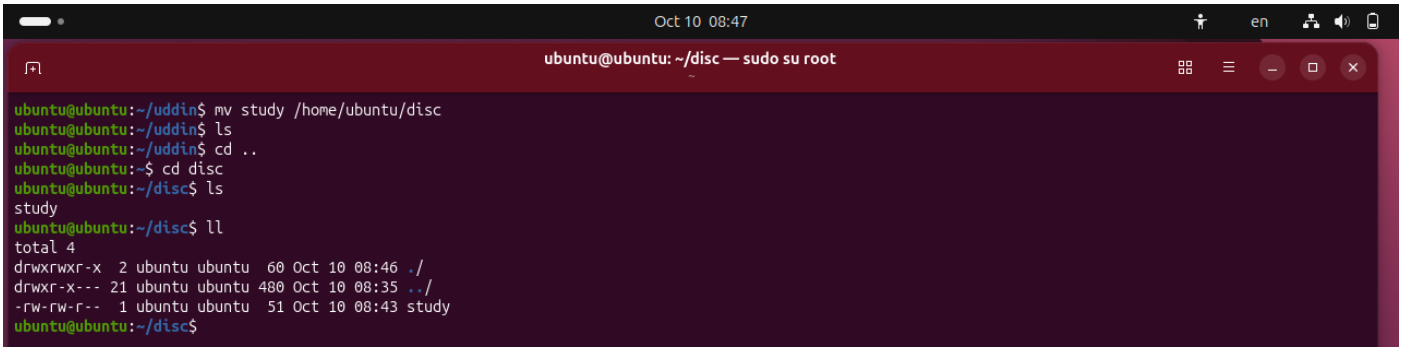
7. **View the contents of a file.**



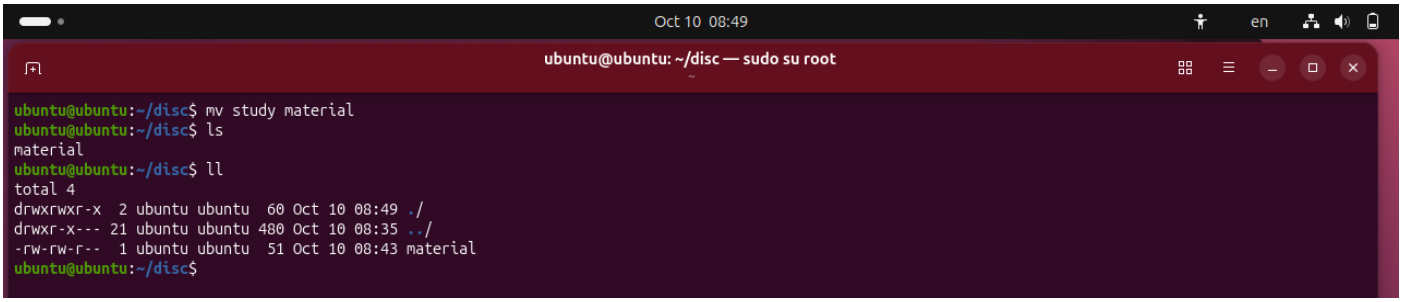
8. **Copy a file to another location.**
   used cp command to copy file to another location
   file was located in viquar dir I copy to Uddin dir



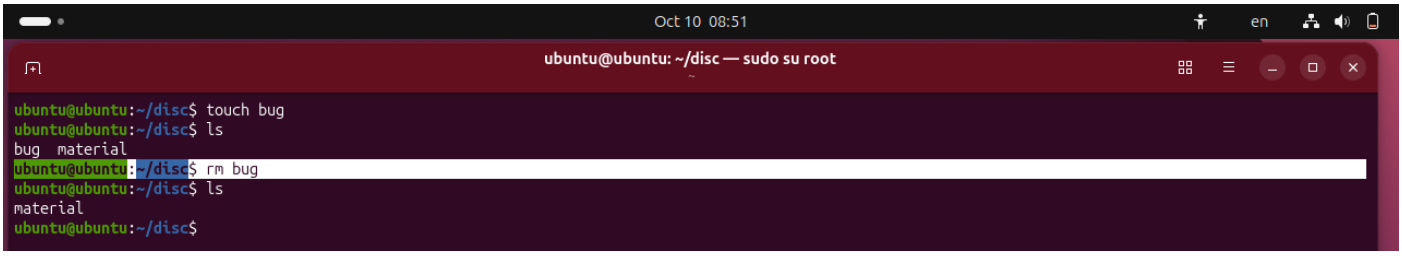
9. **Move a file to another location.**
   used mv command to change location of file
   mv study /home/ubuntu/disc



10. **Rename a file.**
    used mv command to rename file

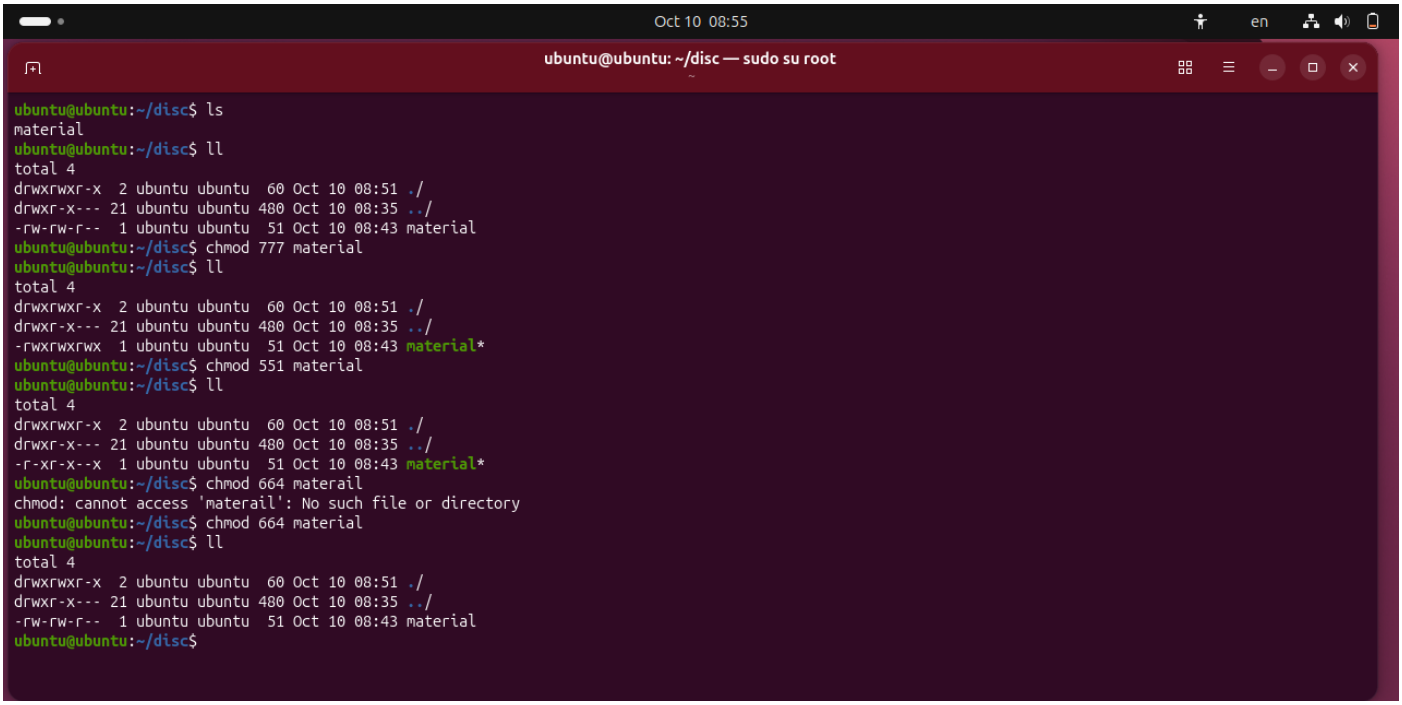


11. **Delete a file.**
    used rm command to remove file



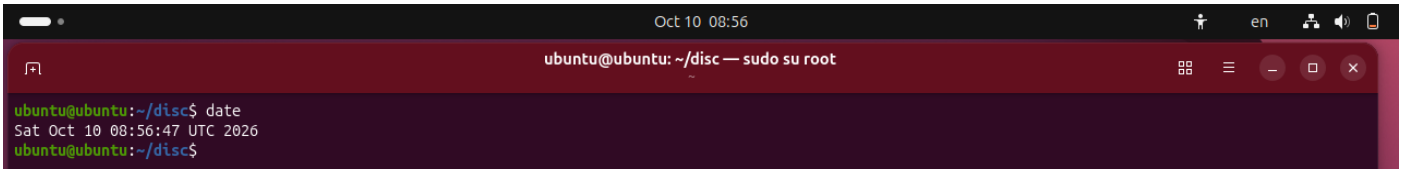
12. Grant or revoke permissions on a file or directory.

used chmod to modify permission of file



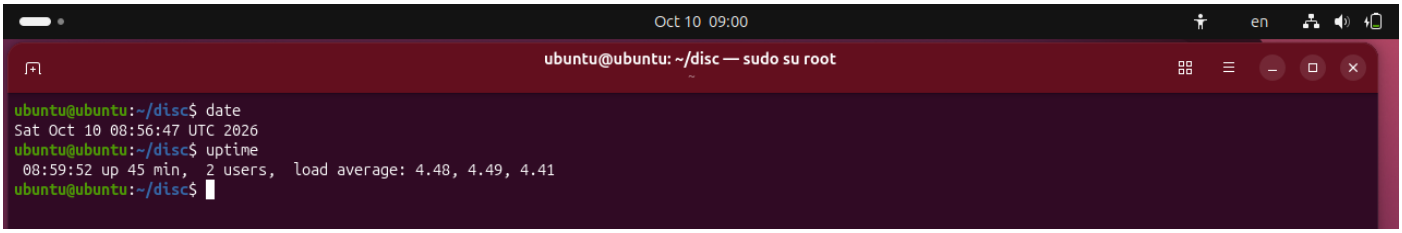
13. View the current date and time.

Used date command



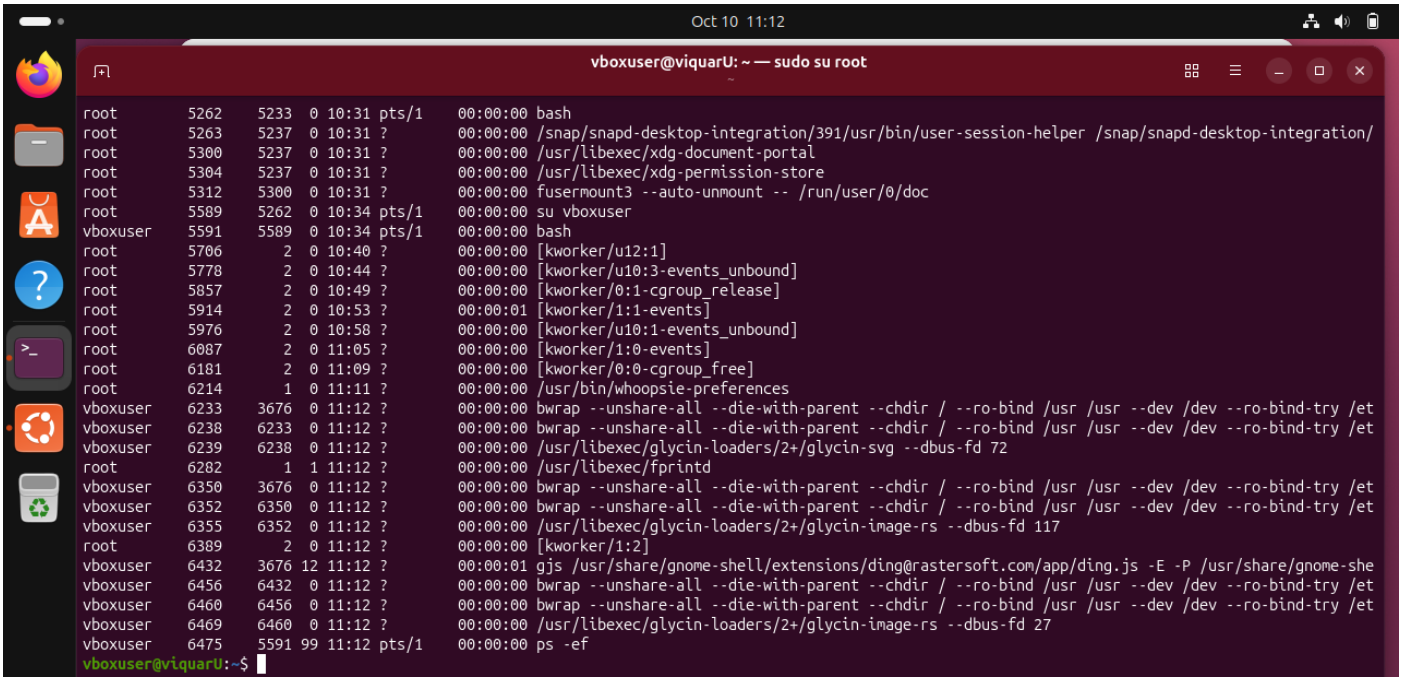
14. Check the system uptime.

used uptime command to see system uptime



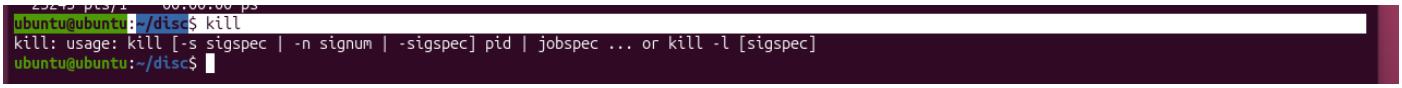
15. View the running processes.

used ps -ef to view running processes and can also use top to check detail running processes



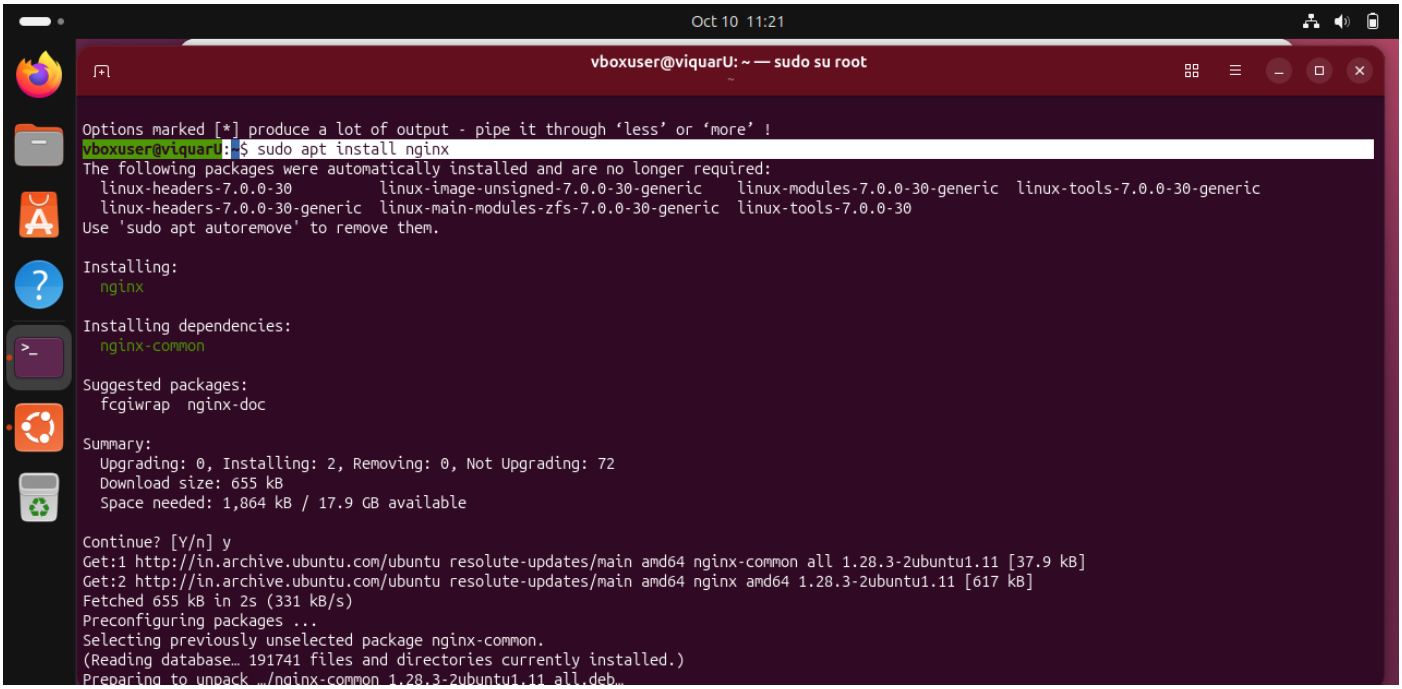
16. Kill a running process.

used kill to kill a running process



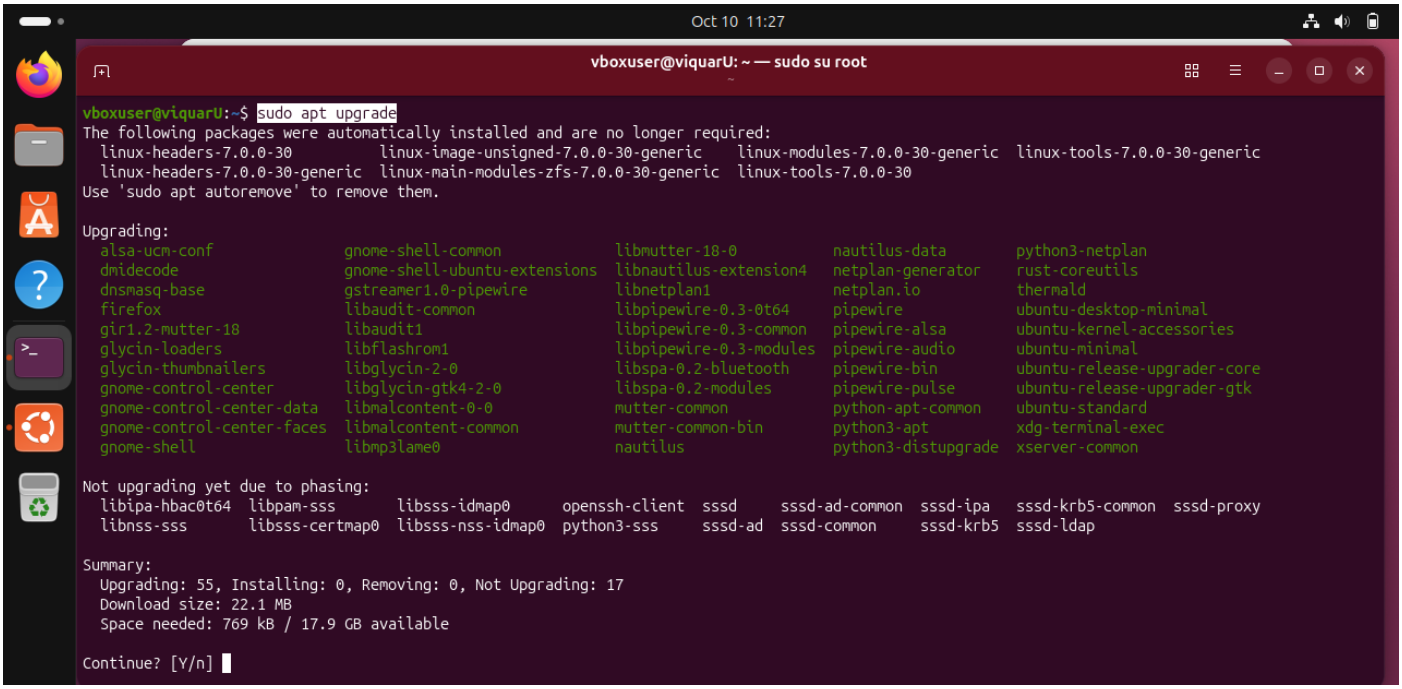
17. Install a package using the package manager (e.g., apt or yum).

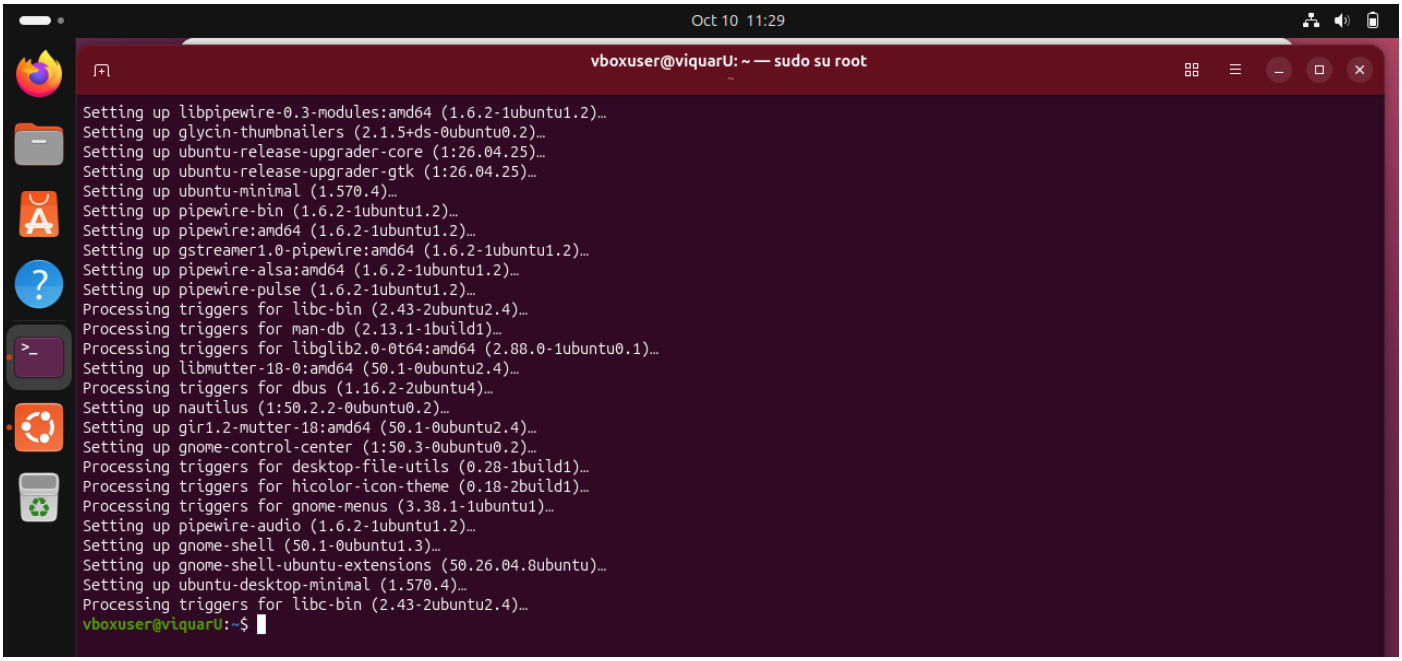
used sudo apt install nginx to install package



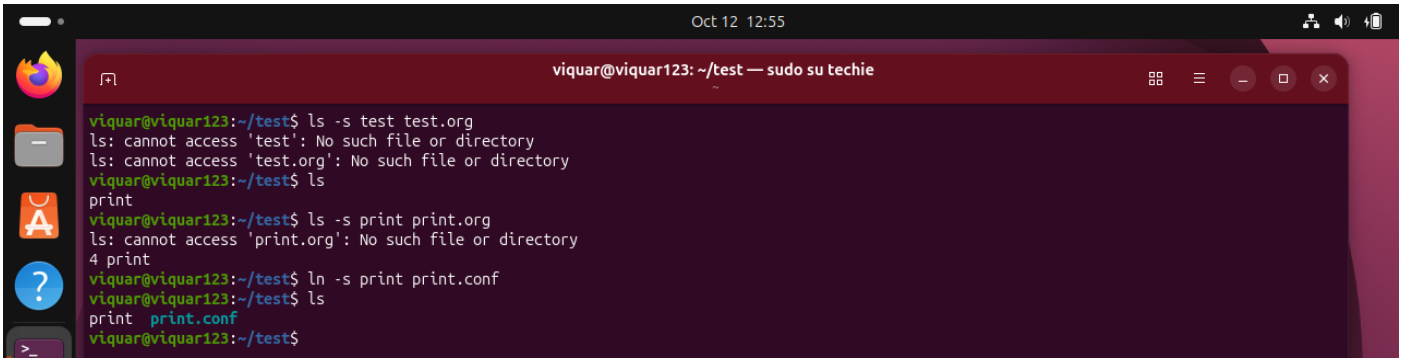
18. Update the system packages.

used sudo apt install upgrade





19. Create a symbolic link.



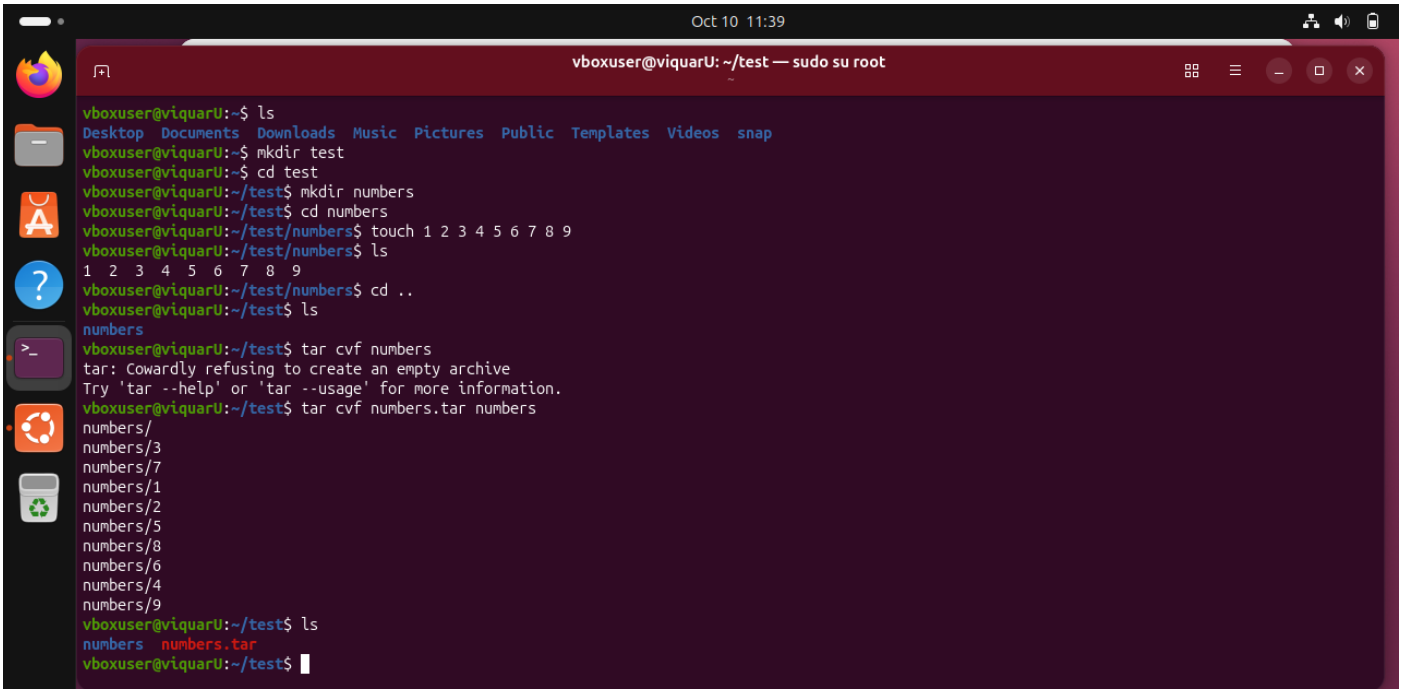
20. Search for files using the find command.

used find / -name (file name) to search file

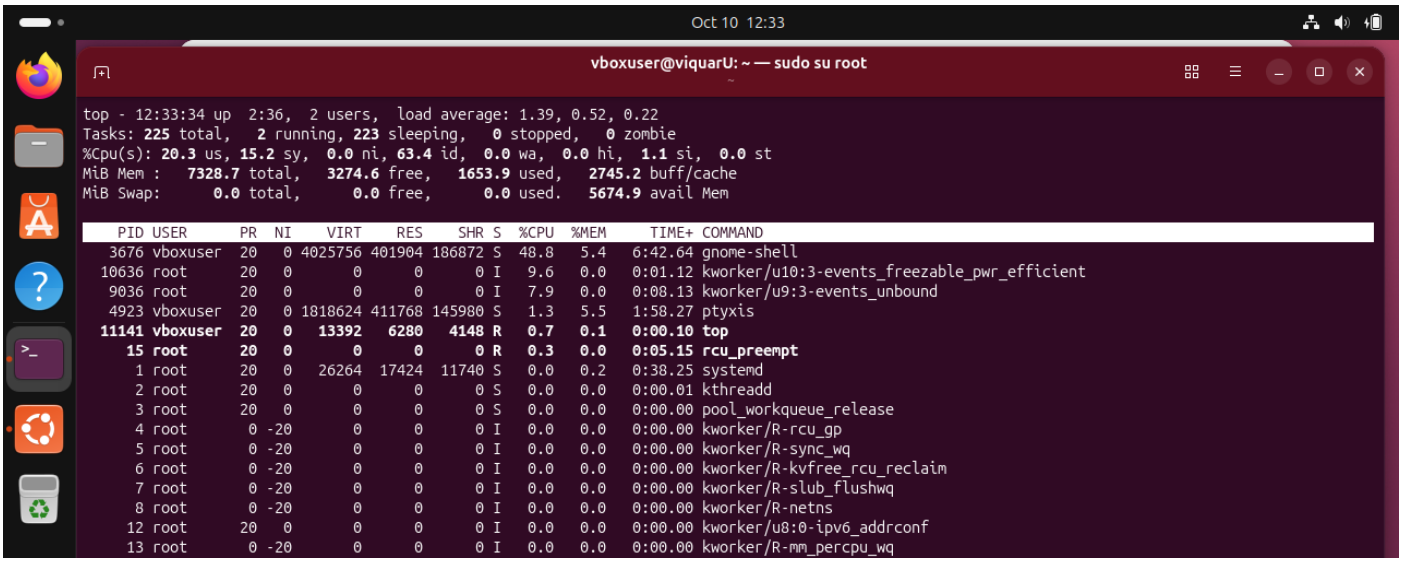


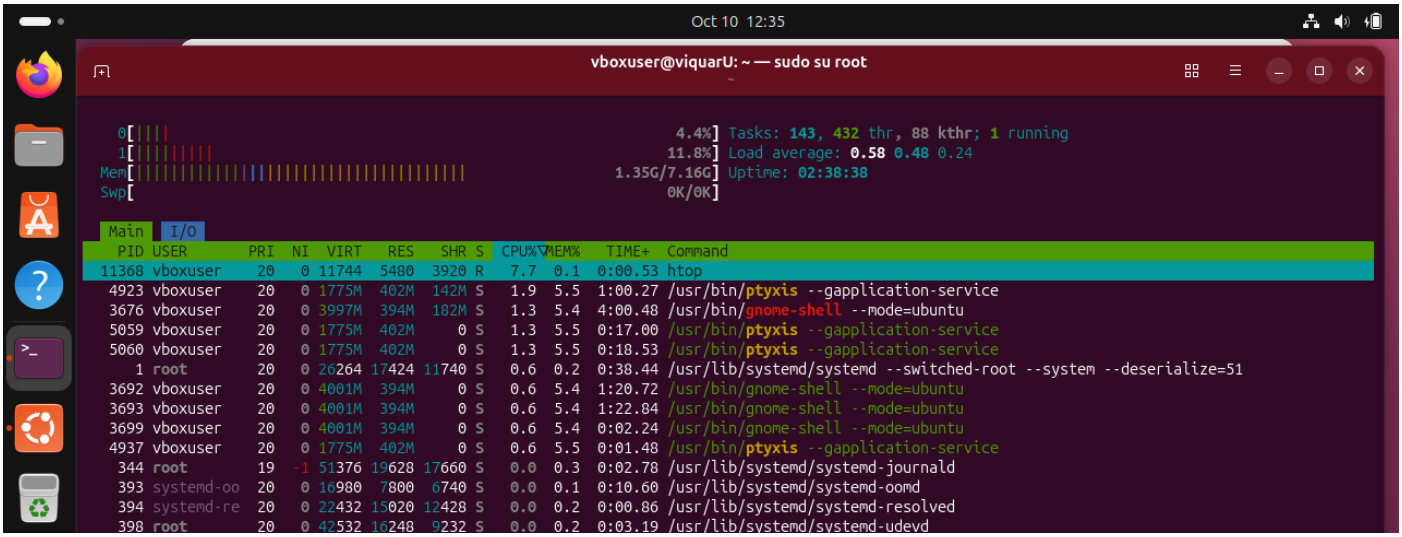
21. Compress and decompress files using tar.

used tar cvf to create .tar file and used tar xvf to extract .tar file



22. Monitor system resources with top or htop.



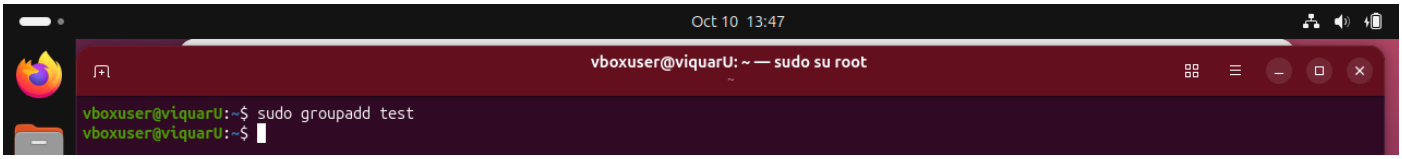


23. Create and manage user groups.

used groupadd command to create group “test”

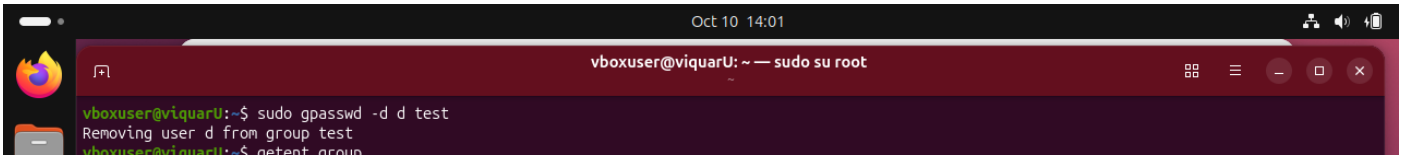
add user a, b, c, & d to test group by using usermod -aG command

remove user from group gpasswd -d username groupname







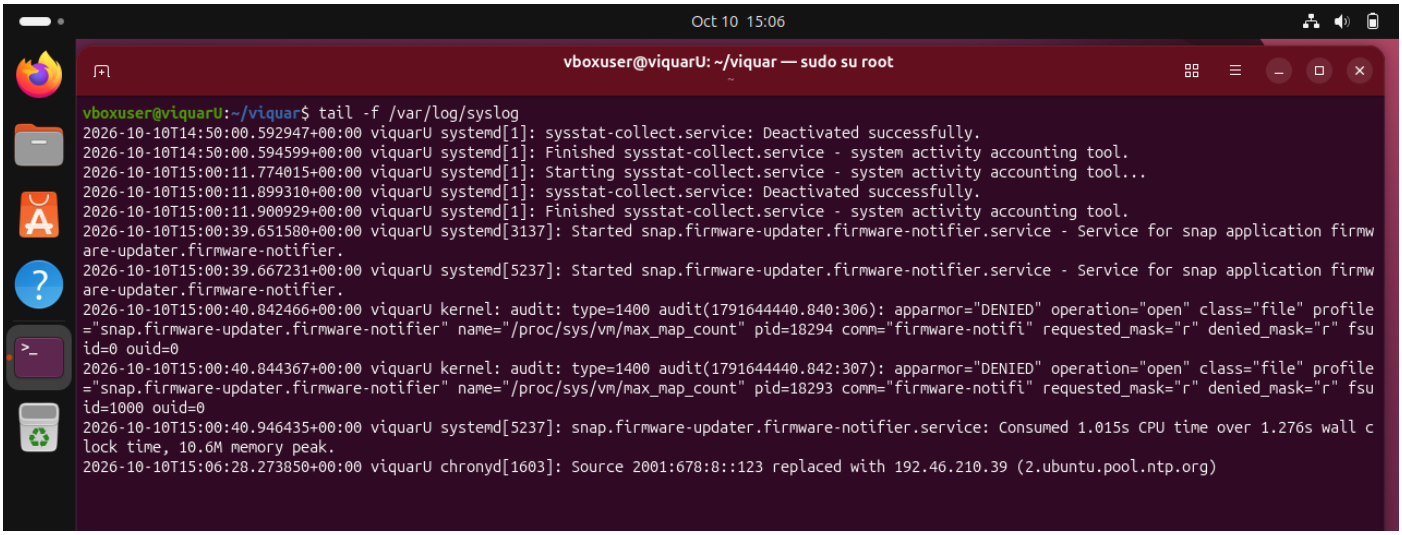




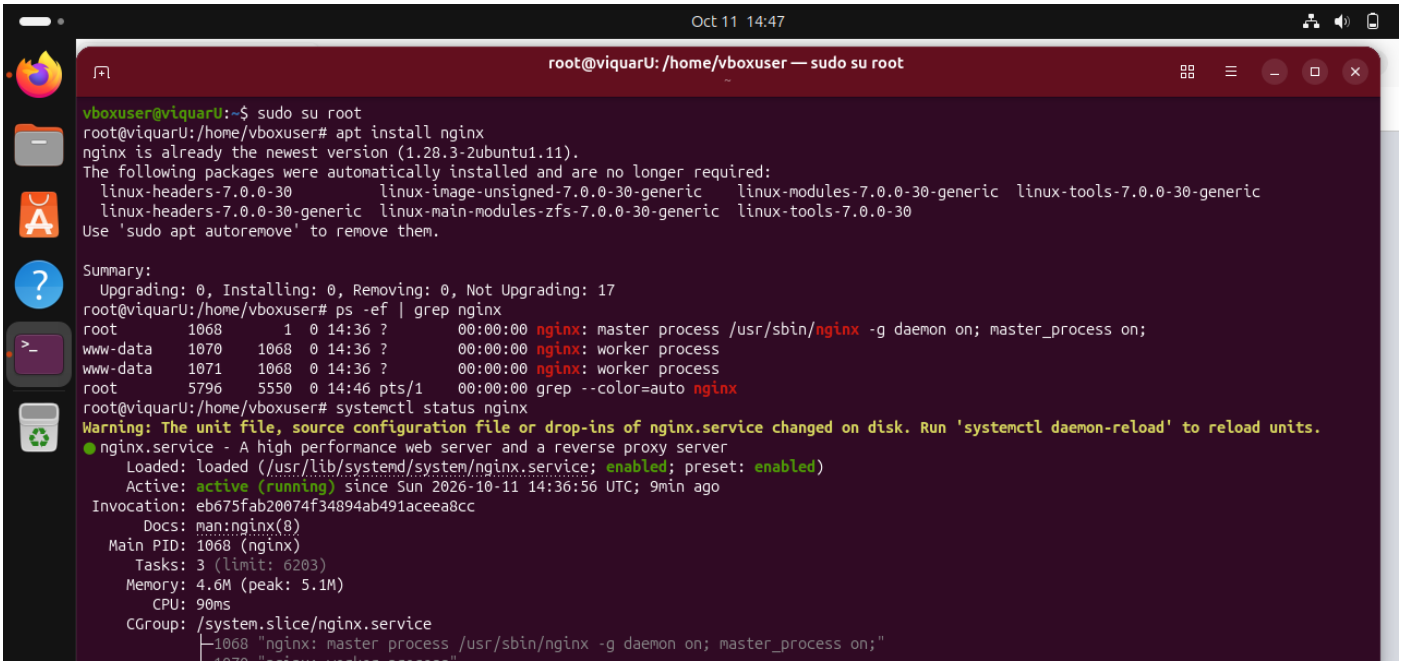
24. Set up SSH password less authentication.

```
HomePC@PC MINGW64 ~/Downloads
$ ssh -i testing-10.pem ec2-user@16.176.137.91
The authenticity of host '16.176.137.91 (16.176.137.91)' can't be established.
ED25519 key fingerprint is SHA256:3IDBUdTB7GEfN3Td1g6zm5OD6YtXQoNG1PmKtSUOHpE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '16.176.137.91' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-29-213 ~]$
```

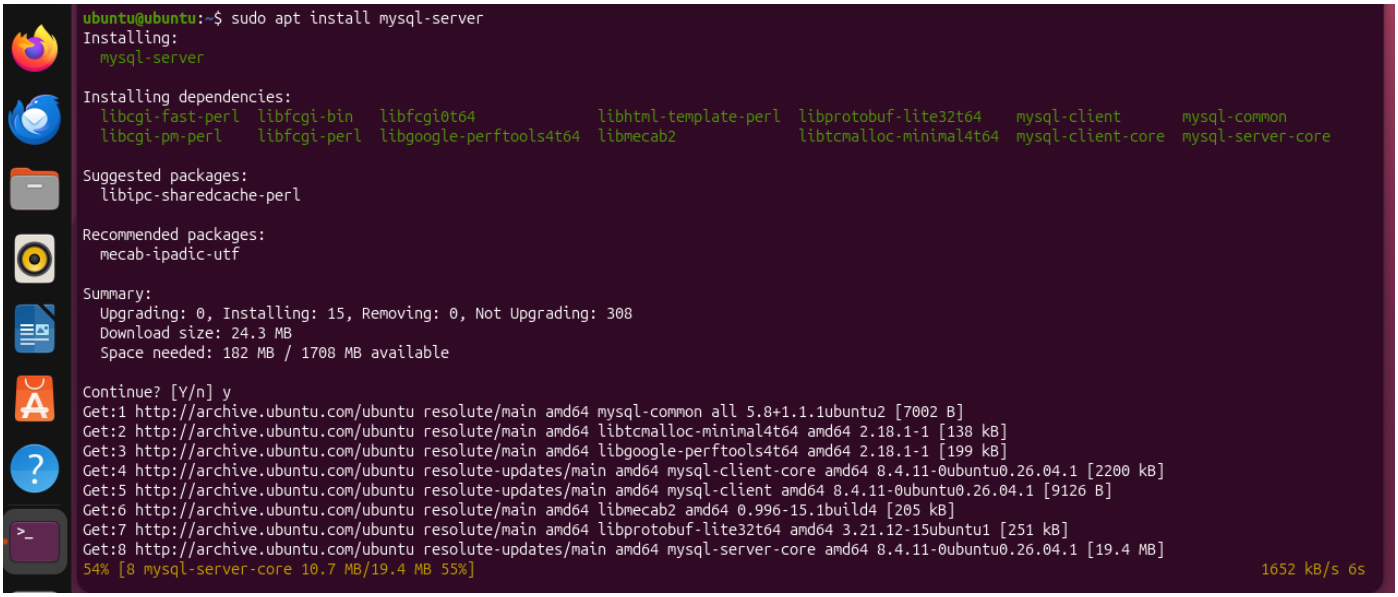
25. Monitor log files using tail or grep.



26. Set up a web server (e.g., Apache or Nginx).



27. Configure and secure a MySQL Database.



```
ubuntu@ubuntu:~$ sudo apt install mysql-server
Installing:
  mysql-server

Installing dependencies:
  libcgi-fast-perl  libfcgi-bin   libfcgi0t64             libhtml-template-perl  libprotobuf-lite32t64    mysql-client       mysql-common
  libcgi-pm-perl    libfcgi-perl  libgoogle-perftools4t64  libmecab2              libtcmalloc-minimal4t64  mysql-client-core  mysql-server-core

Suggested packages:
  libipc-sharedcache-perl

Recommended packages:
  mecab-ipadic-utf

Summary:
  Upgrading: 0, Installing: 15, Removing: 0, Not Upgrading: 308
  Download size: 24.3 MB
  Space needed: 182 MB / 1708 MB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 mysql-common all 5.8+1.1.1ubuntu2 [7002 B]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libtcmalloc-minimal4t64 amd64 2.18.1-1 [138 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libgoogle-perftools4t64 amd64 2.18.1-1 [199 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-client-core amd64 8.4.11-0ubuntu0.26.04.1 [2200 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-client amd64 8.4.11-0ubuntu0.26.04.1 [9126 B]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libmecab2 amd64 0.996-15.1build4 [205 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libprotobuf-lite32t64 amd64 3.21.12-15ubuntu1 [251 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-server-core amd64 8.4.11-0ubuntu0.26.04.1 [19.4 MB]
54% [8 mysql-server-core 10.7 MB/19.4 MB 55%]                                                                    1652 kB/s 6s
```



```
ubuntu@ubuntu:~$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.
The 'validate_password' component is installed on the server.
The subsequent steps will run with the existing configuration
of the component.

Skipping password set for root as authentication with auth_socket is used by default.
If you would like to use password authentication instead, this can be done with the "ALTER_USER" command.
See https://dev.mysql.com/doc/refman/8.0/en/alter-user.html#alter-user-password-management for more information.

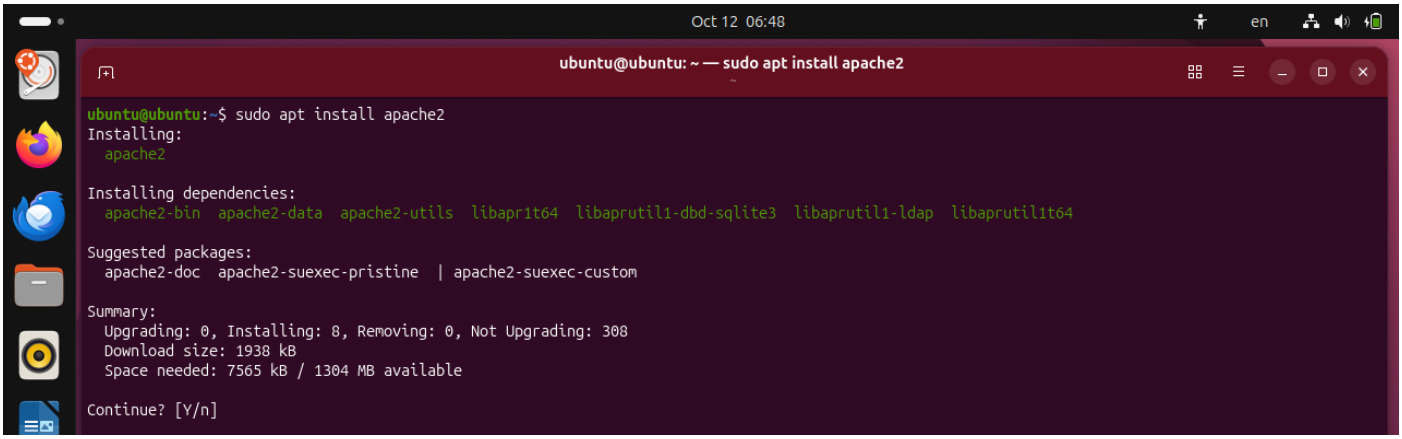
By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
a user account created for them. This is intended only for
testing, and to make the installation go a bit smoother.
You should remove them before moving into a production
environment.

Remove anonymous users? (Press y|Y for Yes, any other key for No) : y
Success.


Normally, root should only be allowed to connect from
'localhost'. This ensures that someone cannot guess at
the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : y
Success.
```

28. Set up a Application Server (e.g.,Apache Tomcat)



```
Oct 12  06:48                                                                    en
ubuntu@ubuntu: ~ — sudo apt install apache2
ubuntu@ubuntu:~$ sudo apt install apache2
Installing:
  apache2

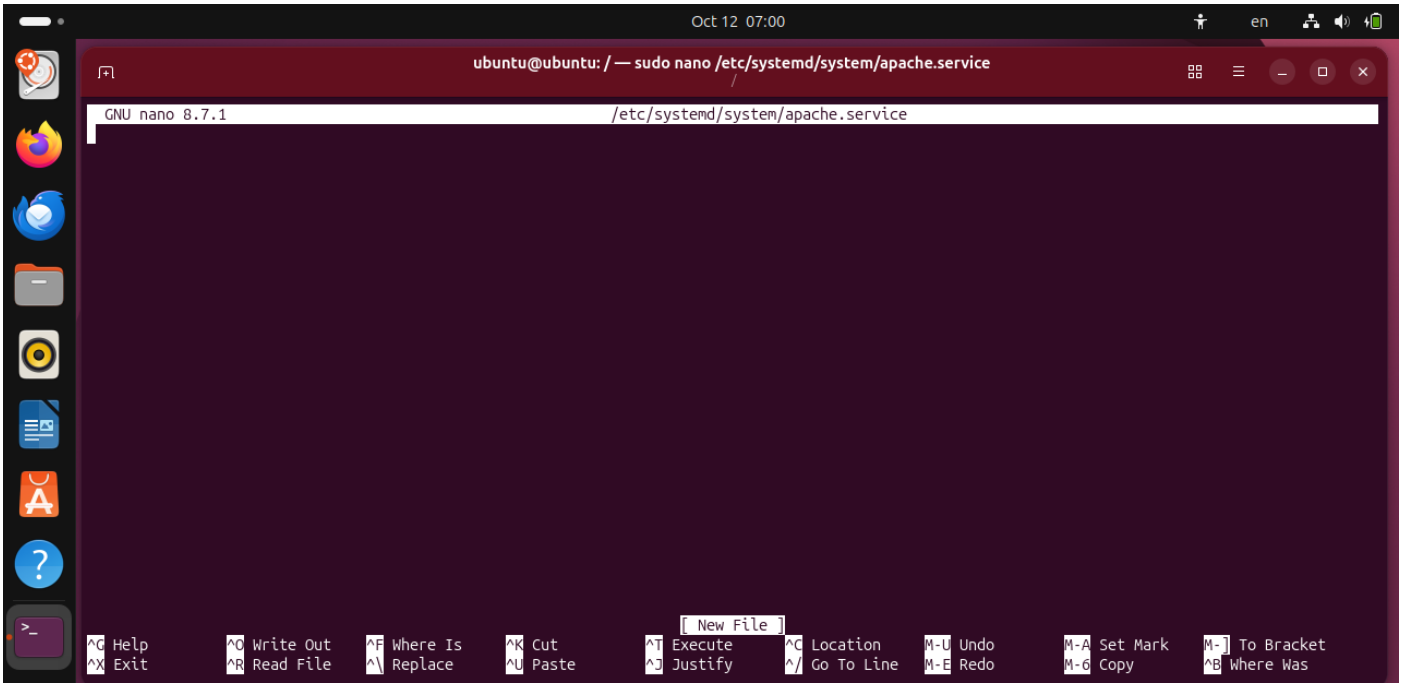
Installing dependencies:
  apache2-bin  apache2-data  apache2-utils  libapr1t64  libaprutil1-dbd-sqlite3  libaprutil1-ldap  libaprutil1t64

Suggested packages:
  apache2-doc  apache2-suexec-pristine  | apache2-suexec-custom

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 308
  Download size: 1938 kB
  Space needed: 7565 kB / 1304 MB available

Continue? [Y/n]
```

29. create a service file for Apache Tomcat.(Should execute by using systemtctl command)



30. Print specific columns from a delimited file.

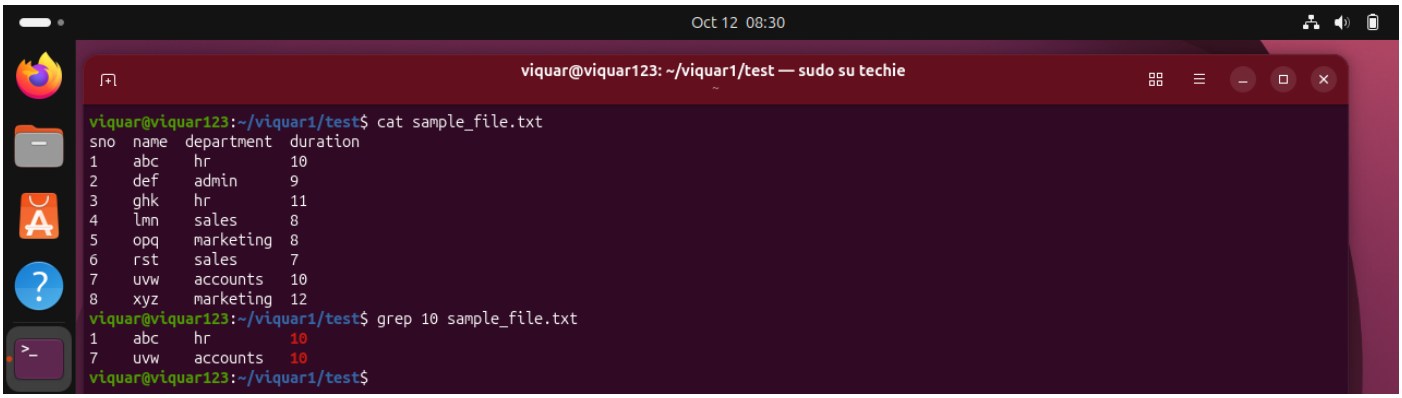
used ps -ef | awk '{print $2}' to print specific columns



31. Filter and print lines based on a specific pattern or condition.

used grep command (grep 10 sample_file.txt) to print files lines based on a specific pattern

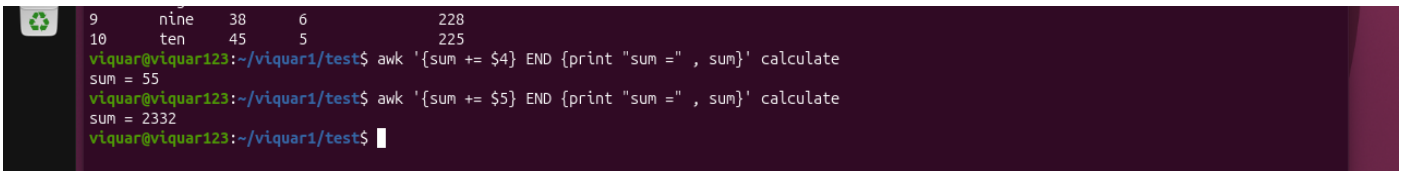
used awk command (awk '4$ > 8) sample_file.txt) to get data of Greter than 10

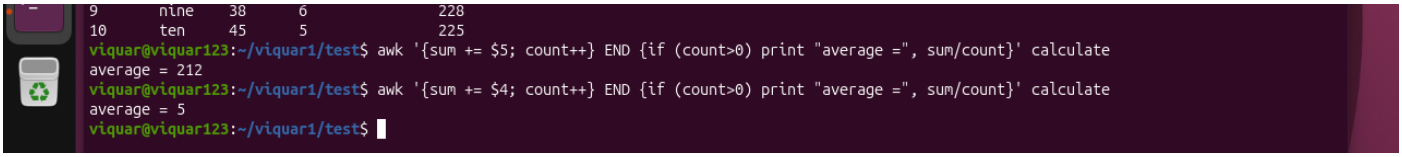


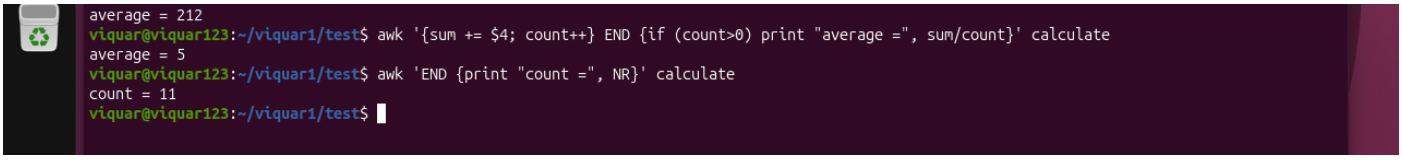


32. **Calculate and print the average, sum, or other statistics of a column.**

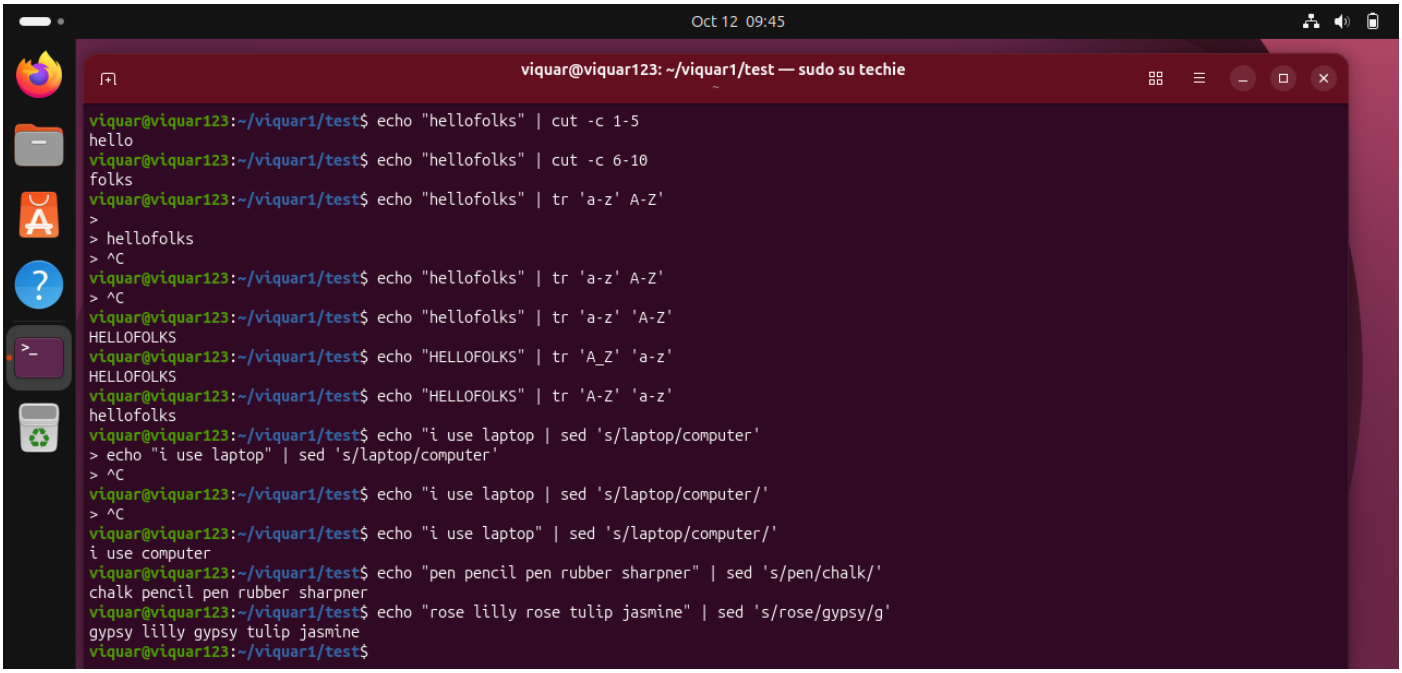
used awk command to sum (awk ‘{sum += $4} END {print “sum =” , sum}’ calculate)
used awk command to sum (awk ‘{sum += $5} END {print “sum =” , sum}’ calculate)
used awk command to sum (awk '{sum += $4; count++} END {if (count>0) print "Average =", sum/count}' Calculate)
used awk command to sum (awk '{sum += $5; count++} END {if (count>0) print "Average =", sum/count}’ calculate)
awk 'END {print "Count =", NR}' calculate

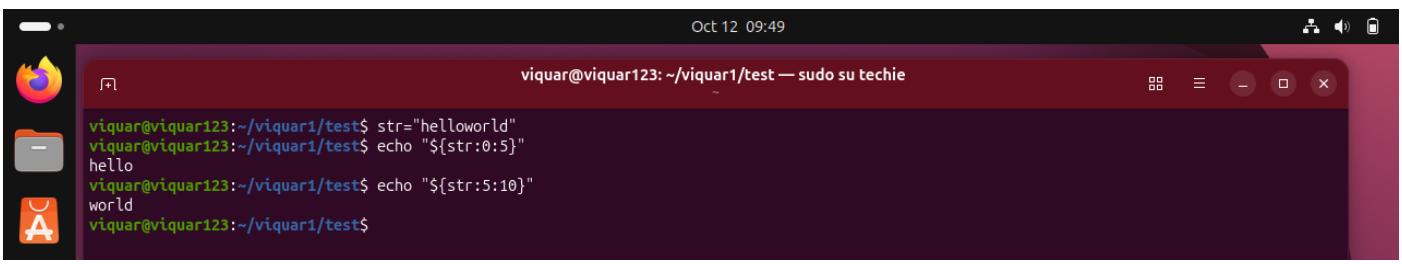






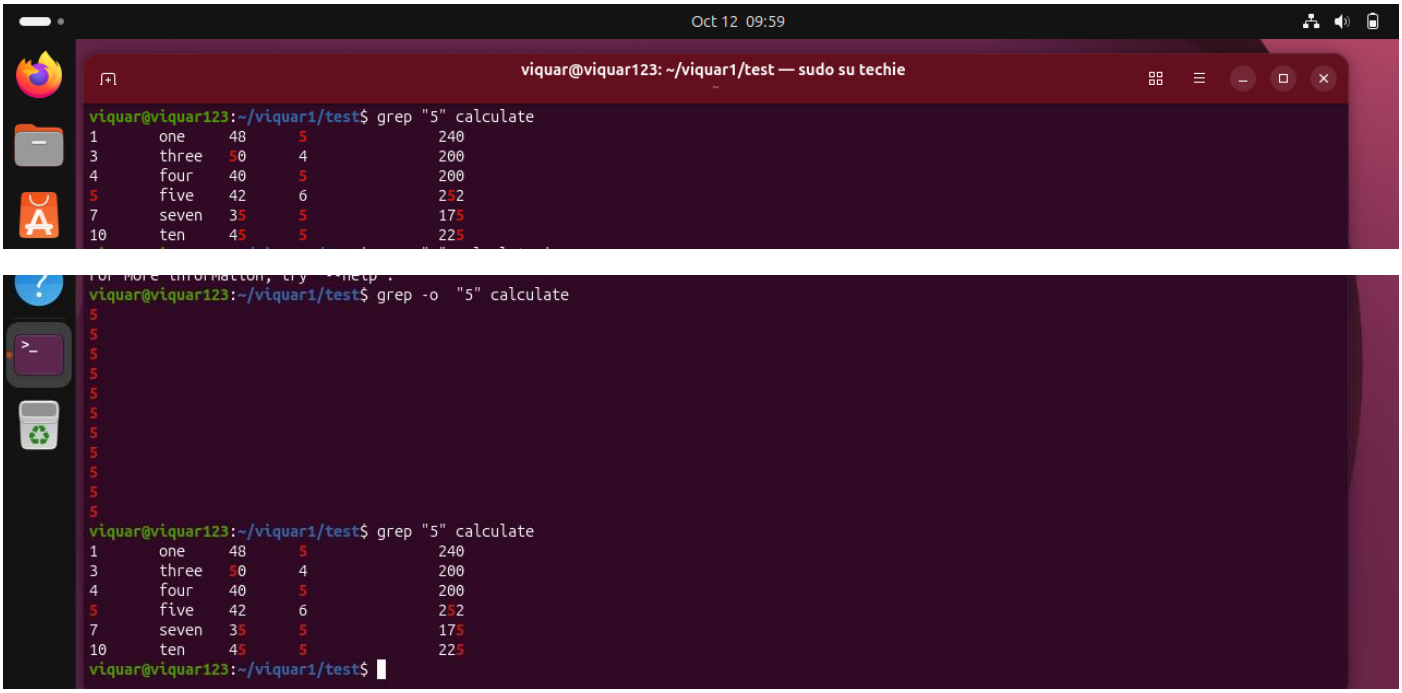
33. **Perform string manipulation, such as extracting substrings or changing case.**





34. Count the occurrences of a specific pattern in a file.

used grep command (grep “5’ calculate) 5 is pattern calculate is file name

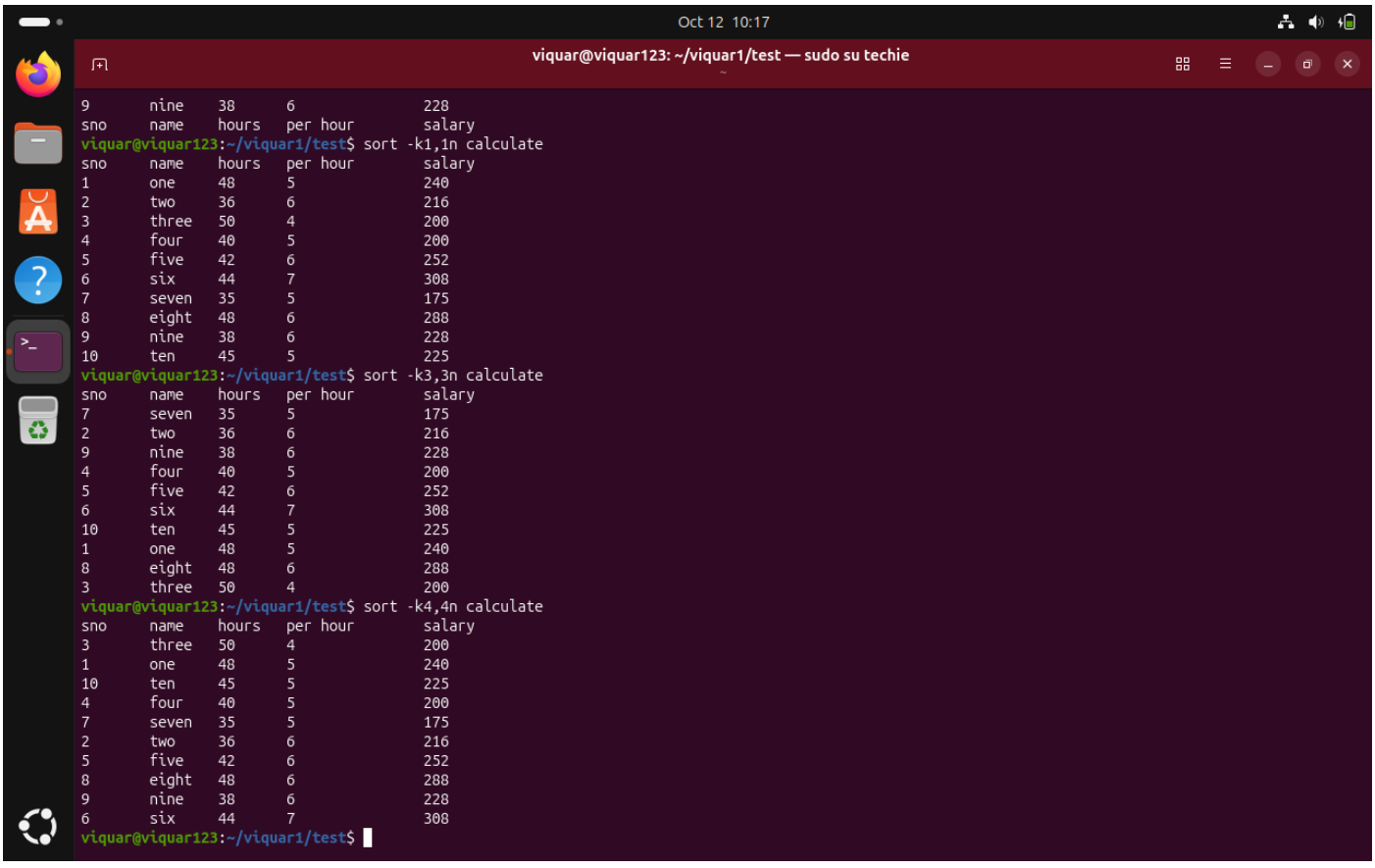


35. Sort lines based on a specific field or column.

used sort command (sort -k1,1n calculate) -k1,1 describe column n described to numerical order and calculate is file name

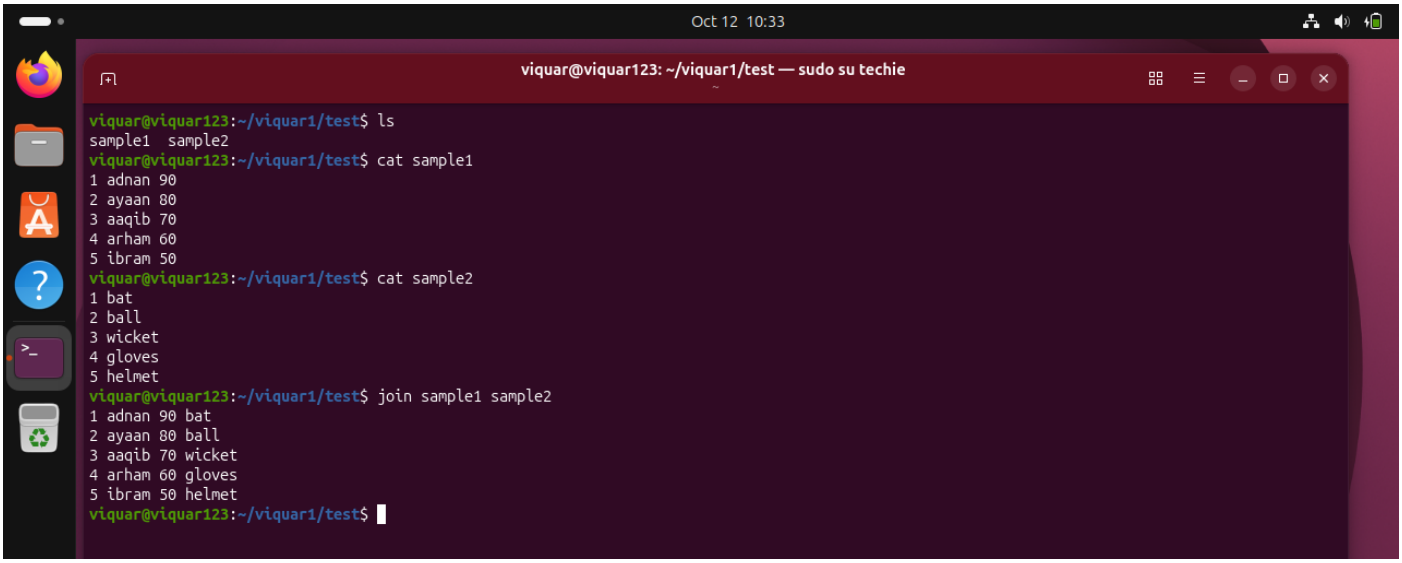
(sort -k1 calculate) -k1 is row and it will arrange alphabetical order

(sort -k1,1nr) for descending order



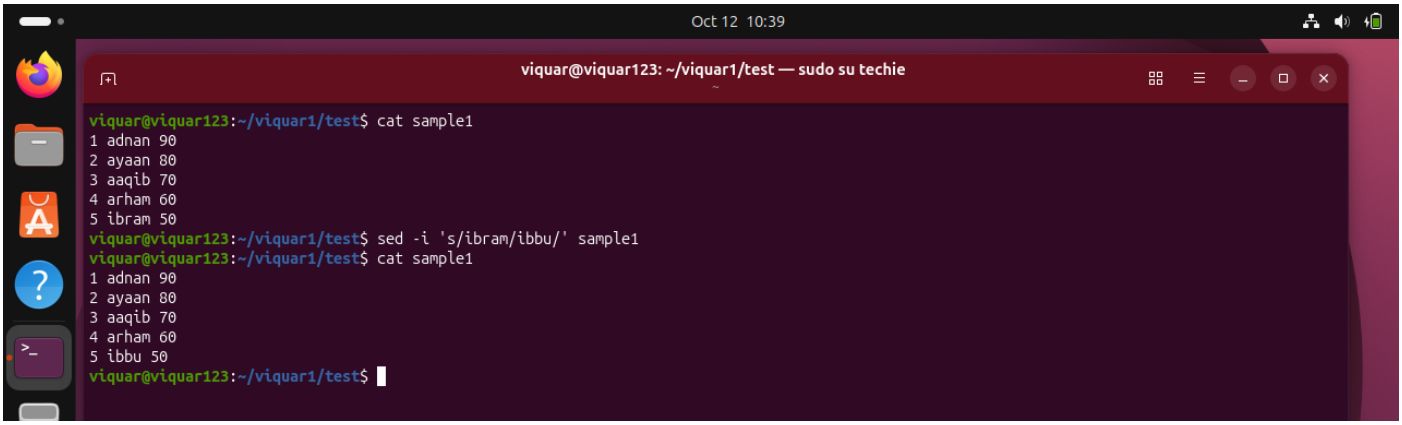
36. Merge multiple files based on a common field or column.

used join command (join sample1 sample2) where sample1 and sample2 are files



37. Substitute text in a file using search and replace.

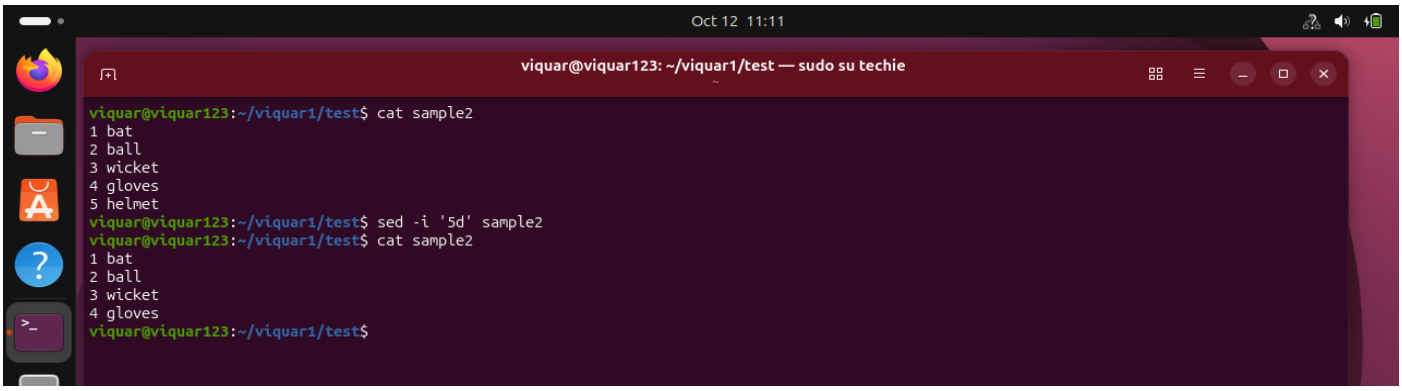
used sed command (sed -i ‘s/ibram/ibbu/’ sample1) -i is make changes in file aswell, if you doesn’t want to change to file just use (sed ‘s/ibram/ibbu/’)

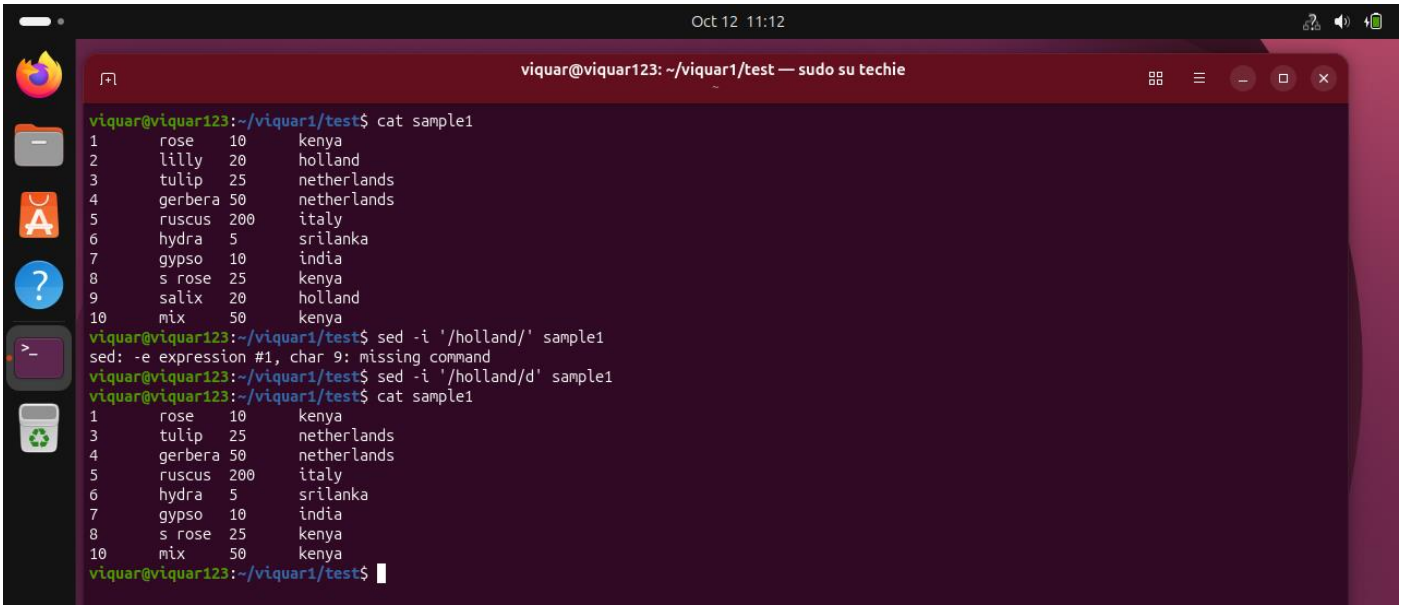


38. Delete specific lines based on a pattern or line number.

used sed command (sed -i ‘5d’ sample2) to delete line without pattern

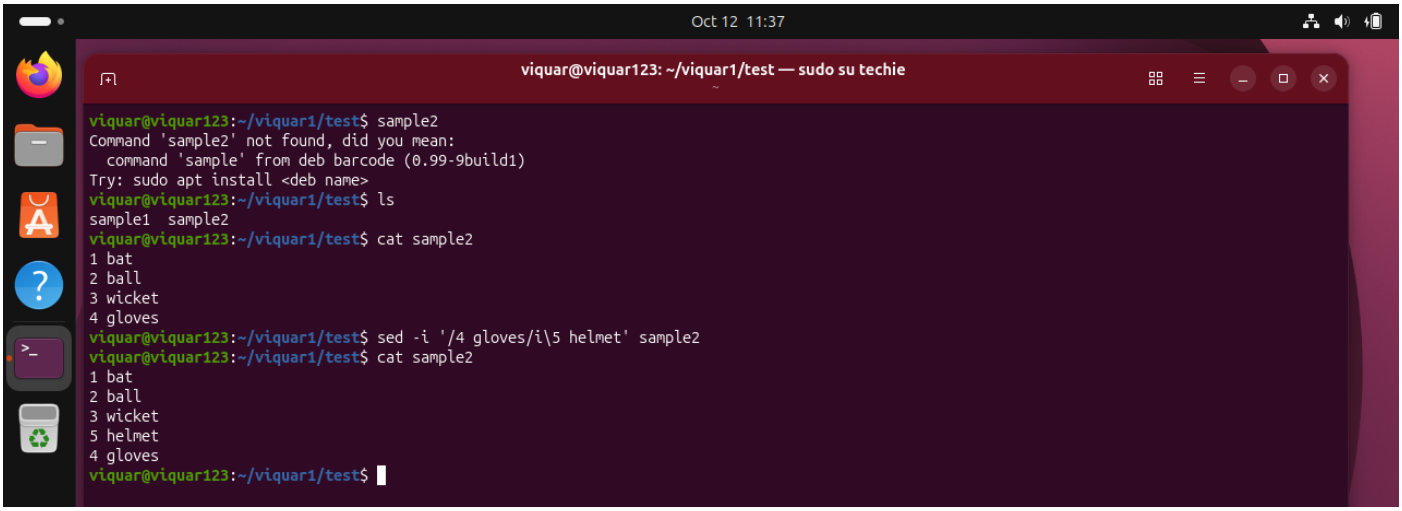
use sed command (sed -I ‘/holland/d’ sample1) to delete line with pattern





39. Append or insert text before or after a specific pattern or line.

used sed command (sed -i '/4 gloves/i\5 helmet' sample2) 4 gloves was existing text and inserted 5 helmet after 4 gloves

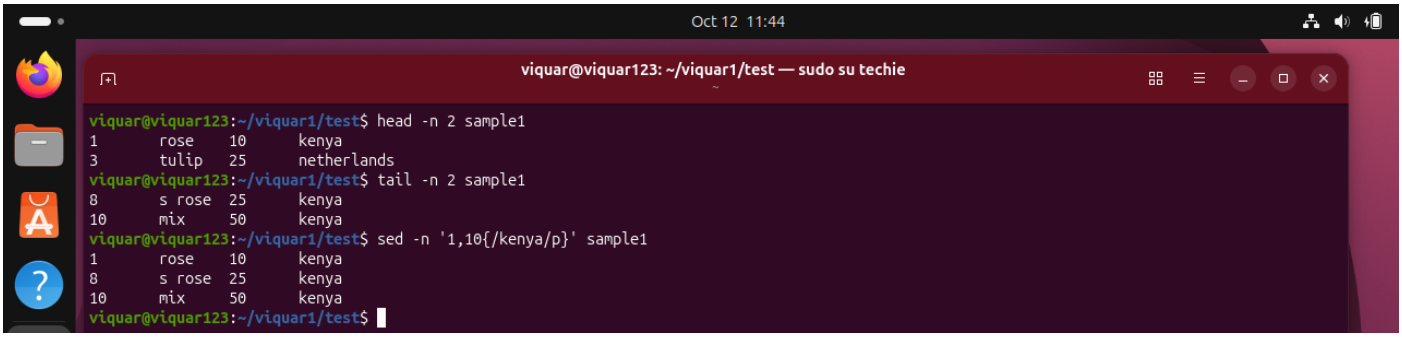


40. Print only specific lines from a file.

used head (head -n 2 sample1) to print first 2 lines

used tail (tail -n 2 sample1) to print last 2 lines

used sed (sed -n '1,10{/kenya/p}' sample1

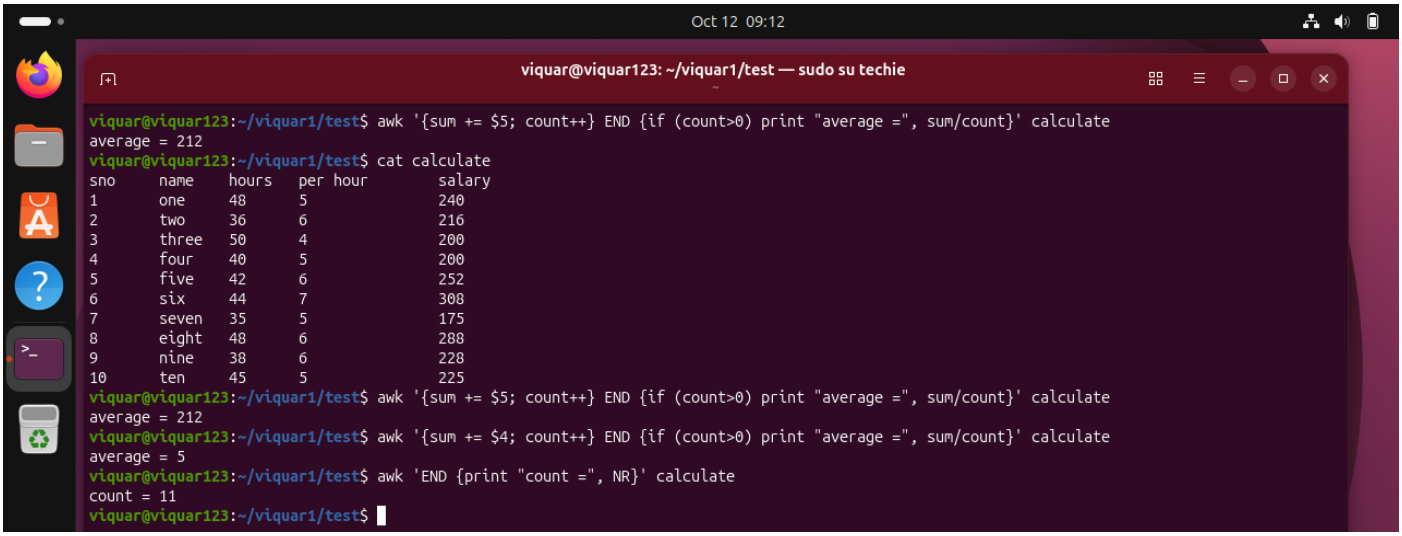


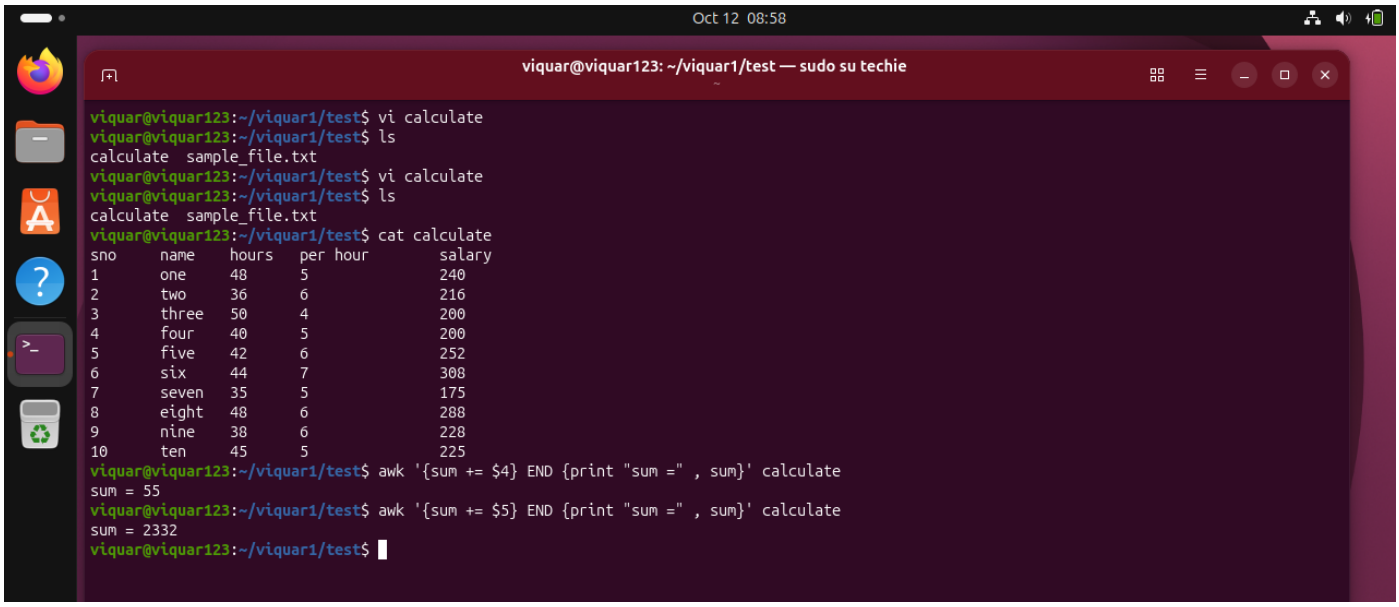
41. Copy file from linux to windows machine

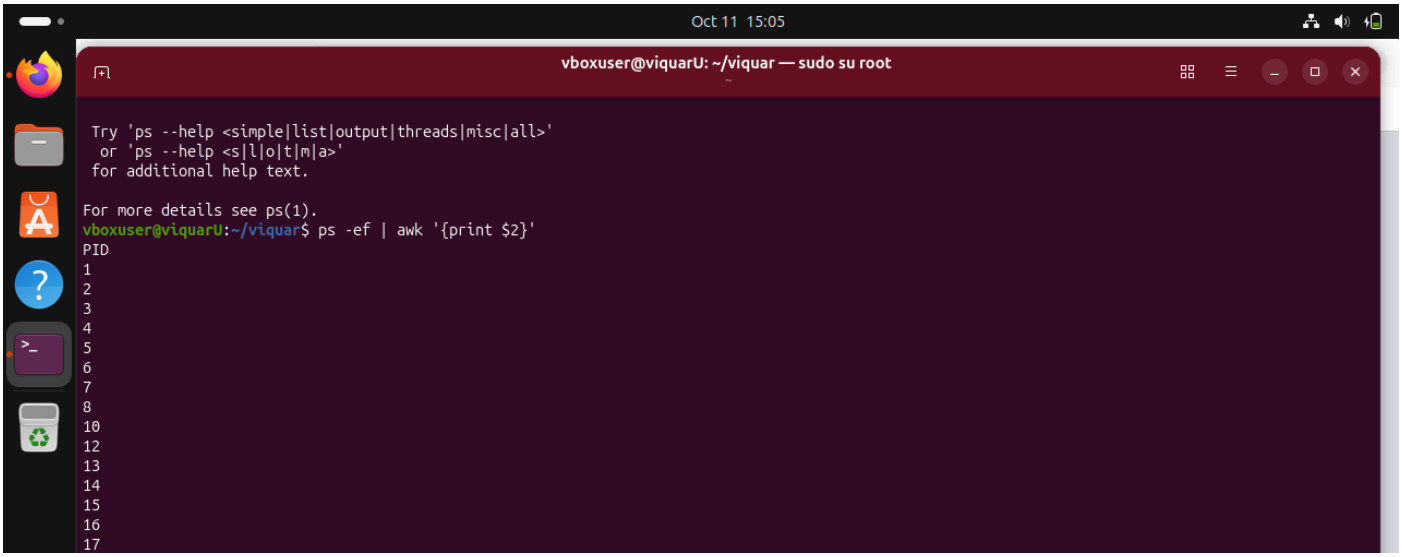
```
HomePC@PC MINGW64 ~/downloads
$ scp -i testing-10.pem ec2-user@16.176.137.91:/home/ec2-user/viquar/linux ~/Downloads/
linux

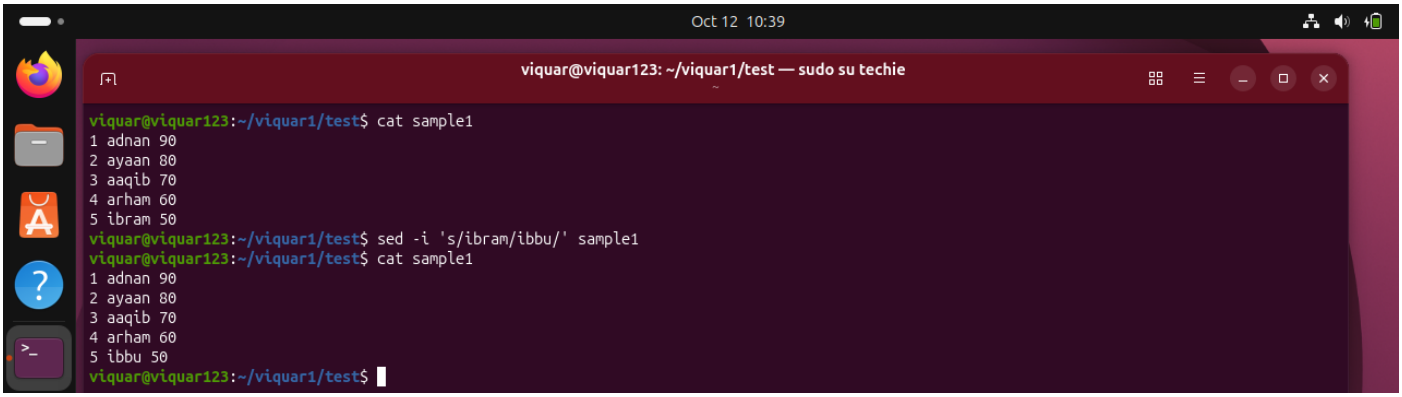
HomePC@PC MINGW64 ~/downloads
$
```

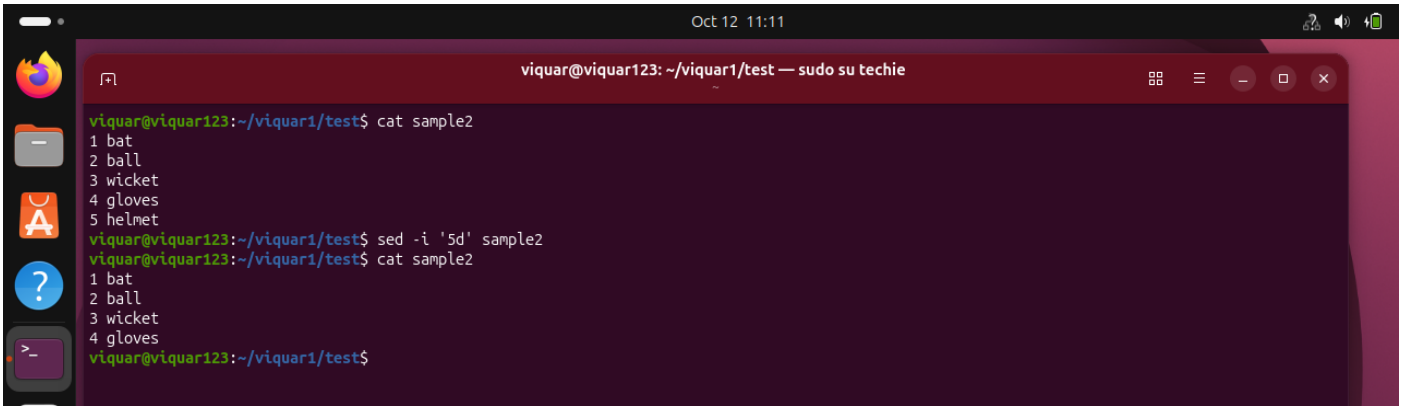
42. 5 use cases for AWK and 5 use cases f or sed

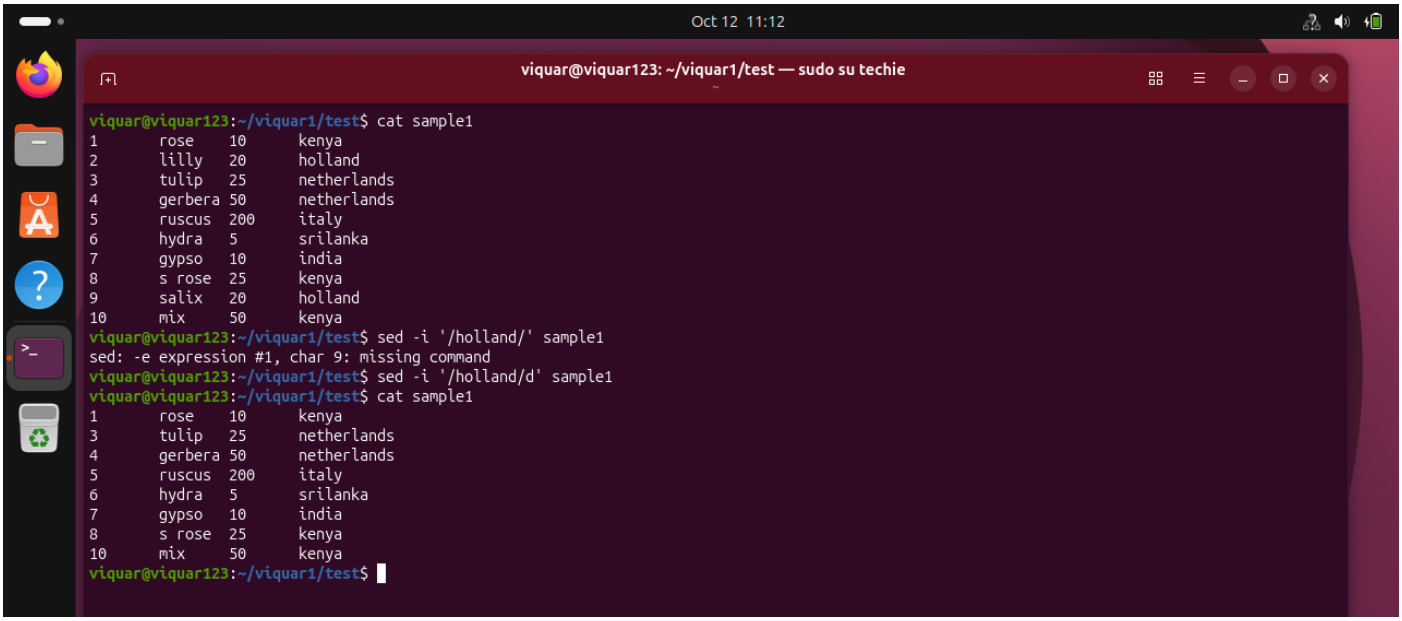


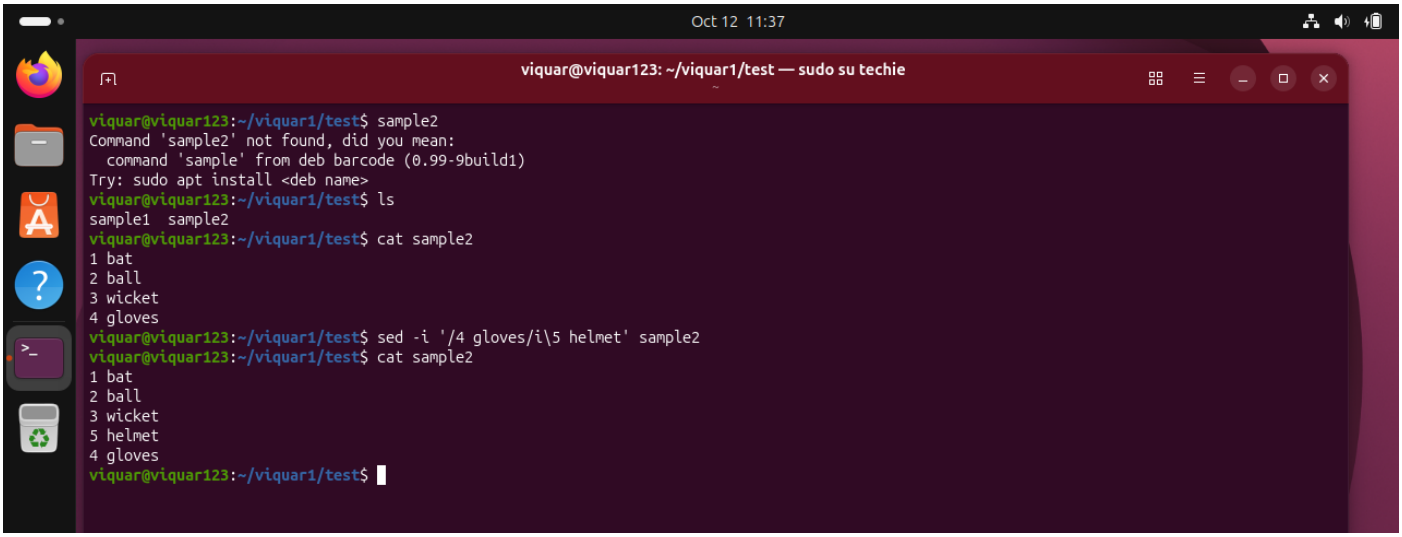



Oct 11 15:05
vboxuser@viquarU: ~/viquar — sudo su root
Try 'ps --help <simple|list|output|threads|misc|all>'
or 'ps --help <s|l|o|t|m|a>'
for additional help text.
For more details see ps(1).
vboxuser@viquarU:~/viquar$ ps -ef | awk '{print $2}'
PID
1
2
3
4
5
6
7
8
10
12
13
14
15
16
17


Oct 12 10:39
viquar@viquar123: ~/viquar1/test — sudo su techie
viquar@viquar123:~/viquar1/test$ cat sample1
1 adnan 90
2 ayaan 80
3 aaqib 70
4 arham 60
5 ibram 50
viquar@viquar123:~/viquar1/test$ sed -i 's/ibram/ibbu/' sample1
viquar@viquar123:~/viquar1/test$ cat sample1
1 adnan 90
2 ayaan 80
3 aaqib 70
4 arham 60
5 ibbu 50
viquar@viquar123:~/viquar1/test$


Oct 12 11:11
viquar@viquar123: ~/viquar1/test — sudo su techie
viquar@viquar123:~/viquar1/test$ cat sample2
1 bat
2 ball
3 wicket
4 gloves
5 helmet
viquar@viquar123:~/viquar1/test$ sed -i '5d' sample2
viquar@viquar123:~/viquar1/test$ cat sample2
1 bat
2 ball
3 wicket
4 gloves
viquar@viquar123:~/viquar1/test$


Oct 12 11:12
viquar@viquar123: ~/viquar1/test — sudo su techie
viquar@viquar123:~/viquar1/test$ cat sample1
1 rose 10 kenya
2 lilly 20 holland
3 tulip 25 netherlands
4 gerbera 50 netherlands
5 ruscus 200 italy
6 hydra 5 srilanka
7 gypso 10 india
8 s rose 25 kenya
9 salix 20 holland
10 mix 50 kenya
viquar@viquar123:~/viquar1/test$ sed -i '/holland/' sample1
sed: -e expression #1, char 9: missing command
viquar@viquar123:~/viquar1/test$ sed -i '/holland/d' sample1
viquar@viquar123:~/viquar1/test$ cat sample1
1 rose 10 kenya
3 tulip 25 netherlands
4 gerbera 50 netherlands
5 ruscus 200 italy
6 hydra 5 srilanka
7 gypso 10 india
8 s rose 25 kenya
10 mix 50 kenya
viquar@viquar123:~/viquar1/test$


Oct 12 11:37
viquar@viquar123: ~/viquar1/test — sudo su techie
viquar@viquar123:~/viquar1/test$ sample2
Command 'sample2' not found, did you mean:
command 'sample' from deb barcode (0.99-9build1)
Try: sudo apt install <deb name>
viquar@viquar123:~/viquar1/test$ ls
sample1 sample2
viquar@viquar123:~/viquar1/test$ cat sample2
1 bat
2 ball
3 wicket
4 gloves
viquar@viquar123:~/viquar1/test$ sed -i '/4 gloves/i\5 helmet' sample2
viquar@viquar123:~/viquar1/test$ cat sample2
1 bat
2 ball
3 wicket
5 helmet
4 gloves
viquar@viquar123:~/viquar1/test$